

Chapter 3: Open Sources

The Source Landscape and How to Work It

Companion Workbook to the Instructional Video

How to Use This Workbook

This is the written companion to the instructional video on open sources. It functions as a follow-along guide, a standalone textbook chapter, and a practice manual.

Chapter 1 established what intelligence is and how the lifecycle works. Chapter 2 established what OSINT is, where it came from, and why verification is the discipline's central skill. **This chapter is about the sources themselves.** Nineteen categories, what each one holds, what each one is good and bad at, and how to choose among them.

That last point is the reason this chapter exists. Beginners fail at open source work not because they cannot search but because they search the wrong category. Asking a social media platform a question that a court docket answers definitively is the single most common waste of investigative time. By the end of this chapter you should be able to look at a question and know, before searching anything, which two or three categories are likely to hold the answer.

Time estimate:

Component	Estimated time
Video	60 to 75 minutes
Reading material	5 to 6 hours
Knowledge assessment	1 hour
Critical thinking exercises	1.5 to 2 hours
Labs 1 through 5	10 to 15 hours
Capstone assignment	5 to 8 hours
Total	24 to 33 hours

Materials: a computer with internet access, a browser, a word processor, and the sample metadata files provided alongside this workbook for Lab 5. No paid tools or subscriptions are required for any exercise.

The Three Rules Still Apply

Carried forward from Chapter 2, enforced in every lab and the capstone:

Rule 1: No private individuals. Every exercise uses fictional entities, organizations, or historical public matters. Researching a real private person as part of this coursework is a zero regardless of the quality of the work.

Rule 2: Publicly available means genuinely public. No false identities, no credential use, no defeating access controls, no accepting material from someone unauthorized to share it.

Rule 3: Document as you go. Source, URL or citation, access date and time, archived copy, and ratings for every finding.

This chapter adds a fourth rule, because it introduces the dark web:

Rule 4: No dark web access in this course. Section 1.18 teaches you what the dark web is, what it holds, why it matters analytically, and how professionals handle it. You will not access it, and no exercise asks you to. The reasons are given in that section and they are good ones.

Learning Objectives

Knowledge objectives (you can explain it):

1. Define an open source and explain what distinguishes a source category from a search tool.
2. Describe the content, access model, coverage, and reliability profile of each of the nineteen categories in this chapter.
3. Distinguish primary, secondary, and tertiary sources and explain why the distinction changes how much a finding is worth.
4. Distinguish structured from unstructured data and explain the analytic consequences of each.
5. Explain the difference between the surface web, the deep web, and the dark web, and correct the common confusion between them.
6. Explain what metadata is, where it comes from, what it can establish, and why it is corroborating rather than conclusive.
7. Explain why archived websites are a distinct source category rather than a convenience.
8. Describe the coverage gaps and biases characteristic of each category.

Application objectives (you can do it):

1. Given an investigative question, select the two or three source categories most likely to answer it, and justify the selection.
2. Classify a set of sources by category, type, and structure.

3. Locate a specific fact in a government record, a court docket, and a corporate filing.
4. Compare archived versions of a website and identify substantive changes.
5. Extract metadata from a file and state what it does and does not establish.
6. Corroborate a claim across categories rather than within one category.
7. Produce a source-attributed intelligence product with reliability and credibility ratings and named gaps.

Analysis objectives (you can judge it):

1. Assess a source's reliability on the basis of its production process rather than its familiarity.
 2. Recognize bias, agenda, and misinformation across categories, including in official sources.
 3. Explain why corroboration within one category is weaker than corroboration across categories.
 4. Determine when a question cannot be answered from open sources and say so.
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Part 1: Reading Material

1.1 Introduction to Open Sources

1.1.1 A Category Is Not a Tool

A **source category** is a class of information defined by who produces it, through what process, and under what obligations. A **tool** is a means of searching one or more categories.

This distinction matters more than it sounds. A search engine is a tool that indexes part of one category (the surface web) and almost none of several others. Students who think of the internet as a source rather than as a delivery channel end up believing that anything not in the first two pages of results does not exist. The most valuable material in most investigations sits in categories that general search engines index poorly or not at all: court dockets behind case-management systems, regulatory filings in agency portals, archived pages no longer live, print holdings never digitized, and structured databases that require a query rather than a keyword.

The practical consequence. When you plan collection, plan by category and then pick tools. Never the reverse.

1.1.2 The Six Properties That Describe Any Source

Every category in this chapter is described using the same six properties. Learn the framework now and the nineteen sections become variations on a theme rather than nineteen things to memorize.

Property	The question it answers	Why it matters
Content	What does this category actually hold?	Prevents asking a category for something it never contains

Property	The question it answers	Why it matters
Production process	Who makes it, under what obligation, with what review?	This is the real basis of reliability, not familiarity
Access model	Free, fee, registration, in person, restricted?	Determines cost and time in your collection plan
Coverage	What jurisdictions, dates, languages, and populations are included, and which are missing?	Coverage gaps produce false negatives, the most dangerous error in OSINT
Latency	How long between the event and the record's availability?	A source with 90-day latency cannot answer a question due Friday
Failure mode	How does this category typically mislead a careless analyst?	Each category misleads in a characteristic way

Production process is the key column. A corporate filing is reliable not because companies are honest but because the filing is made under legal penalty, on a schedule, with defined content requirements, and is subject to review. A social media post is unreliable not because people lie constantly but because there is no process at all: no obligation, no schedule, no defined content, no review. Reliability is a property of process, not of medium or reputation.

1.1.3 Primary, Secondary, and Tertiary

Carried in from research methodology, and it is the single most useful classification in this chapter.

Primary sources are original, first-hand records created at or near the time by a participant, a system, or an official process. A court filing, a deed, an SEC annual report, a photograph taken at the scene, raw sensor data, a transcript of testimony, the text of a statute.

Secondary sources describe, interpret, or summarize primary material. A news article about the court filing. An academic paper analyzing the annual reports of forty companies. A documentary about the event.

Tertiary sources compile and index secondary material. An encyclopedia entry. A textbook. A database that aggregates news summaries.

Why this determines the value of a finding. Every step away from the primary source adds a chance for error, compression, and interpretation. A news article stating that a company "faces a \$2.1M lawsuit" may have misread the docket, may be describing a claim rather than a judgment, or may be summarizing a different case. The docket says exactly what it says.

The working rule: always walk the claim back to the primary source before relying on it. If a secondary source cites a filing, get the filing. If it cites a study, get the study. Most analytic errors in open source work are secondary-source errors, and most are avoidable by one additional retrieval step.

The important exception. Primary is not automatically better for every purpose. A primary source is a data point; a good secondary source may supply the context, the comparison, and the prior history that

a single data point cannot. And primary sources have their own biases: a police report is a primary source that records one participant's account, and a corporate filing is a primary source written to satisfy a regulator. Primary means *closer to the event*, not *true*.

1.1.4 Structured and Unstructured

Structured data is organized into defined fields with consistent types: a spreadsheet of permits, a court docket's case metadata, a filing's financial statement tables, vessel position feeds.

Unstructured data is free-form: article text, video, audio, forum posts, photographs, scanned documents.

Semi-structured data sits between: a document with consistent headings, an email with headers and a free-text body, an HTML page with predictable markup.

Why it matters practically:

	Structured	Unstructured
Search method	Query by field	Keyword, then human reading
Comparability	High; can be aggregated and counted	Low; requires extraction first
Volume tolerance	Very high	Low; humans must read it
Common error	Trusting a field without checking its definition	Misreading meaning, or missing what was never said
Typical categories	Databases, filings, dockets, registries, maps, sensor feeds	News, books, social posts, video, audio, imagery

The single most valuable habit here: **when you can convert an unstructured question into a structured one, do it**. "Has this company been sued" is an unstructured search across news. "List all cases in these four jurisdictions where this entity or its former names appear as a party" is a structured query, and it produces a defensible answer with a stated coverage boundary.

1.1.5 A Map of the Nineteen Categories

OPEN SOURCE LANDSCAPE

PUBLISHED / EDITED

Books Newspapers Academic papers
Radio Television Podcasts
(Process: editorial or peer review. Secondary mostly.)

OFFICIAL / COMPELLED

Government records Court records Corporate filings
Financial records Public databases
(Process: legal obligation. Primary, structured.)

OBSERVATIONAL / TECHNICAL

Satellite imagery Maps Metadata
(Process: instrument or system generated. Primary.)

USER-GENERATED

Social media
(Process: none. Primary in form, unverified in fact.)

CHANNEL / ACCESS LAYERS (not content categories)

Internet Archived websites Dark web
(These are how you reach content, and each holds material the others do not.)

The last box is deliberately separated. The internet, web archives, and the dark web are not content types; they are access layers that carry many content types. Treating them as categories alongside "court records" is a conceptual error that leads directly to poor collection planning.

1.2 The Internet

1.2.1 What It Is, Precisely

The internet is a **delivery channel**, not a source category. Almost every other category in this chapter is partially reachable through it. Treating it as a source is like treating "the post office" as a source.

The useful decomposition is by accessibility layer:

Layer	Definition	Examples	Reachable by general search?
Surface web	Pages a search engine can crawl and index	News sites, company sites, blogs, public documents linked from pages	Yes
Deep web	Content behind a query, a form, a login, or a	Court case management systems, library	No; requires going to the source and querying

Layer	Definition	Examples	Reachable by general search?
	paywall, or excluded from crawling	catalogs, most databases, subscription archives, agency search portals, webmail	it
Dark web	Content requiring specific overlay network software to reach	Sites on anonymizing networks	No

The proportion point, stated carefully. You will encounter claims that the deep web is hundreds of times larger than the surface web. Treat specific multipliers with caution; they derive from old studies and are not measurable with precision. The defensible version of the claim is the one that matters analytically: **the majority of the most authoritative material is in the deep web, because authoritative material tends to live in databases that must be queried rather than on pages that can be crawled.** That is why a search engine is a starting point and never a finishing point.

1.2.2 The Six Properties

Property	Assessment
Content	Everything, unevenly
Production process	None inherent; varies entirely by what is being delivered
Access model	Mostly free at the surface; fee, registration, or query for much of the deep web
Coverage	Biased toward recent, English-language, commercially valuable, and connected populations
Latency	Minutes to hours at the surface; varies by system for the deep web
Failure mode	Believing that what a search engine returns is what exists

1.2.3 Search Behavior That Separates Beginners From Practitioners

Search engines are ranked, personalized, and commercial. Results are ordered by relevance models optimized for typical users, influenced by your history and location, and shaped by parties with commercial incentives to rank. Two people searching the same phrase get different results. Two searches a month apart get different results. Therefore: record your query, the engine, the date and time, and the result you used, and never treat "I could not find it" as "it does not exist."

Operators still matter. Exact-phrase quoting, site restriction, file-type restriction, and date-range restriction change results dramatically. The specific syntax varies by engine and changes over time, so

learn the concepts rather than memorizing strings: restrict to a domain, restrict to a document format, restrict to a period, require an exact phrase, exclude a term.

Use more than one engine, including a non-Western one. Different crawlers index different material, and regional engines cover regional content that global engines miss.

Go to the authoritative system directly. If you want a court record, go to the court's system. If you want a filing, go to the regulator's portal. If you want a permit, go to the county. A search engine is a way to find out *which* system to query, not a substitute for querying it.

Document negative results. "Searched the state corporate registry for these six name variants on this date; no records returned" is a finding with real value. "I could not find anything" is not.

1.3 Books

1.3.1 Why Books Still Matter in a Chapter Full of Databases

Books are the category students skip, and skipping them produces a characteristic failure: an investigation with excellent recent coverage and no historical depth. Three things books do that nothing else does as well.

Depth on a bounded subject. A book-length treatment of an industry, a region, a company, or a technical field carries context that no article contains.

Coverage of periods before digitization. For anything before roughly the mid-1990s, books and print periodicals may be the only substantive record. Digital archives thin out rapidly as you go back.

Named sourcing you can follow. A serious nonfiction book has endnotes. Those endnotes are a curated list of primary sources someone else already located, which is the cheapest research assistance available anywhere.

1.3.2 The Six Properties

Property	Assessment
Content	Long-form treatment: history, biography, technical reference, industry analysis, memoir, local and institutional history
Production process	Varies enormously. Academic press with peer review, trade press with editing and fact-checking, and self-published with no external review at all
Access model	Purchase, library lending, interlibrary loan, digitized full-text search with limited display, and out-of-copyright full text
Coverage	Strong historically; weak on anything recent, because publishing latency is a year or more
Latency	12 to 24 months typical from writing to

Property	Assessment
	publication
Failure mode	Treating any bound volume as authoritative; self-published and advocacy titles look identical to peer-reviewed ones on a shelf

1.3.3 Reading a Book as an Investigator

You are almost never reading the whole thing. The professional pattern:

1. **Search inside it.** Large-scale book digitization projects allow full-text search across millions of volumes with snippet display, even where the full text is not viewable. That is enough to determine whether a book contains what you need and on which page.
2. **Read the notes first.** The endnotes tell you what the author actually had access to. A book about a company whose notes cite only press releases is a different object from one whose notes cite depositions and internal documents.
3. **Check the publisher and the author's basis.** University press, established trade imprint, advocacy organization, or self-published? What was the author's access?
4. **Use the index as a lookup table.** Then read the three pages that matter.
5. **Extract the primary sources and go get them.** The book is a finding aid. The cited deposition, filing, or archive collection is the primary source you actually want.

1.3.4 Where to Find Them

Libraries remain underused by open source researchers. Practical points: public and university libraries provide free access to subscription databases you would otherwise pay for; interlibrary loan can retrieve nearly any published volume; special collections and local history rooms hold material that exists nowhere online; and reference librarians are, in effect, free professional research consultants. Any student who has not used a reference librarian for a hard research question has left a resource on the table.

1.4 Newspapers

1.4.1 Two Very Different Things Under One Name

Newspapers serve two distinct investigative functions, and conflating them causes errors.

Current reporting is a secondary source about recent events. It is fast, uneven, and frequently derivative.

Newspaper archives are among the most valuable primary-adjacent historical resources in existence. A local paper from 1953 is a contemporaneous record of who owned what, who died, who was

charged, what was built, what closed, who married whom, and what the community thought. For historical work, archived local newspapers are often the only detailed record.

1.4.2 The Six Properties

Property	Assessment
Content	Events, obituaries, legal notices, property transfers, court proceedings, business openings and closures, local government actions, advertisements
Production process	Editorial selection, varying degrees of fact-checking, correction practice at established outlets, and none of the above at content farms
Access model	Free with limits, subscription, library-provided archive databases, microfilm at libraries, and free historical newspaper collections
Coverage	Excellent for places with a surviving local paper; catastrophic gaps where local papers have closed, which is a large and growing problem
Latency	Hours to days for current reporting
Failure mode	Circular reporting; and treating a wire story reprinted in twenty papers as twenty sources

1.4.3 The Legal Notices Section

Worth its own subsection because it is the most underrated open source in this chapter. Statutory publication requirements mean many jurisdictions compel publication in a newspaper of record for: foreclosure notices, probate and estate notices, name changes, fictitious business name filings, government procurement solicitations, tax delinquency lists, zoning and variance hearings, liquor and other license applications, abandoned property notices, and dissolution notices.

This is compelled disclosure carried in a searchable medium. It routinely contains material that appears nowhere else in a form a member of the public can retrieve.

1.4.4 Reading Current Reporting Carefully

Four checks, applied to every article you rely on:

Who is the reporter, and were they there? A byline with a datelined location is different from an aggregated rewrite.

What is the stated sourcing? "According to a filing" is checkable. "According to officials" is not. "According to reports" means the outlet is repeating someone else and should be treated as a citation, not a source.

Is this original or derivative? Wire services supply the same story to hundreds of outlets. Twenty

appearances of one wire story is one source.

What does the correction record look like? An outlet that issues visible corrections has a process. One that never corrects anything either never errs or does not check.

1.5 Academic Papers

1.5.1 What They Offer and What They Cost

Academic literature gives you method, measurement, and a defensible basis for a claim about how something generally works. When you need to say "facilities of this type fail at approximately this rate" or "this technique detects this condition with this accuracy," the academic literature is where a defensible number comes from.

The cost is latency and specificity. A peer-reviewed paper may be two years behind the work it describes, and it will almost never tell you anything about the specific entity you are investigating.

1.5.2 The Six Properties

Property	Assessment
Content	Empirical studies, methods, literature reviews, technical standards work, theses and dissertations, conference proceedings, preprints
Production process	Peer review at journals, program committee review at conferences, and no review at all for preprints
Access model	Subscription journals, open access journals, institutional repositories, preprint servers, author copies, and library access
Coverage	Strong on studied phenomena; systematically weak on anything commercially sensitive, recent, or unfundable
Latency	Months to years from research to publication
Failure mode	Citing a preprint as though it were peer reviewed, and citing a single study as though it were settled science

1.5.3 How to Assess a Paper Without Being a Specialist

You do not need domain expertise to make several useful judgments:

- **Was it peer reviewed, and where?** Journal, conference proceedings, preprint server, or a predatory venue that publishes for a fee without review. Check whether the venue is indexed in mainstream scholarly indexes.

- **How large and how relevant is the sample?** A finding from twelve subjects in one country is not a general law.
- **Is it a single study or a review?** A systematic review or meta-analysis of many studies is worth far more than any one study.
- **Has it been cited, and how?** Citation counts indicate attention, not correctness, and a highly cited paper may be cited because it is disputed. Read a few citing papers to find out which.
- **Who funded it, and is there a declared conflict?** Declared conflicts are normal and not disqualifying. Undeclared ones are the problem.
- **Has it been retracted?** Retraction databases exist and checking takes seconds. Retracted papers continue circulating for years.

Theses and dissertations deserve special mention. They are long, exhaustively sourced, freely available in institutional repositories, and frequently cover narrow local or industrial subjects that no other published work addresses. For a question about a specific region, industry, or facility type, a dissertation is often the single best document in existence.

1.6 Government Records

1.6.1 The Largest and Most Underused Category

Government records are, for most investigations, the highest-value open source category that exists. The reason is process: they are created under legal obligation, on schedules, with defined content, by parties who face consequences for misstating them, and they are frequently required by law to be public.

Students underuse them for three reasons, all fixable. They do not know what exists. They assume access is harder than it is. And the records live in the deep web, so search engines do not surface them.

1.6.2 What Exists, by Level of Government

Level	Record types
Local (city, county)	Property deeds and transfers, assessments, building permits, code enforcement and violations, business licenses, zoning and variance applications, planning commission and council minutes and packets, procurement solicitations and awards, restaurant and facility inspections, vital records (with access limits), voter registration (varies widely by state), utility districts, jail rosters, incident logs
State	Corporate registrations and annual reports, UCC filings, professional licensing and disciplinary actions, insurance filings, environmental permits

Level	Record types
	and violations, workplace safety enforcement, campaign finance and lobbying disclosures, legislative records and bill histories, state court administrative records, motor vehicle records (heavily restricted), state contracts
National	Securities filings, patents and trademarks, aviation and vessel registries, immigration and customs statistics, trade and import records, safety recalls and investigations, environmental releases and enforcement, occupational safety enforcement, federal contracts and grants, sanctions and debarment lists, census and statistical products, regulatory dockets and rulemaking comments, inspector general reports, legislative hearings and testimony
International / multilateral	Trade statistics, development project documents, treaty registries, standards bodies, sanctions lists, international organization reports and audits

1.6.3 The Six Properties

Property	Assessment
Content	See table above; enormous breadth, structured, primary
Production process	Legal obligation, defined forms, agency review, penalties for false statement. The strongest reliability profile of any category
Access model	Free portals, fee-per-document, in-person inspection, and formal records requests. Highly variable by jurisdiction
Coverage	Complete within jurisdiction and scope; nonexistent outside it. Digitization thins rapidly before the 1990s
Latency	Days for some filings, months for inspection reports, years for some statistical products
Failure mode	Assuming a record's absence means the event did not happen, when it may mean the jurisdiction does not require the record, has not digitized it, or files it under a name you did not search

1.6.4 The Records Request as a Collection Method

Most jurisdictions have a statutory public records mechanism: the federal Freedom of Information Act in the United States, state public records acts, and equivalents in many other countries. This is a formal, lawful collection channel available to any member of the public, and it is a legitimate OSINT technique.

Practical guidance:

- **Be specific.** Requests for "all documents relating to X" get denied as overly broad or take years. Requests naming a document type, a date range, and a custodian get filled.
- **Ask for the index first.** Many agencies will provide a list of record series or a log of prior requests, which tells you what to ask for next.
- **Check prior releases.** Agencies frequently post previously released records, and reading requests. Someone may already have obtained what you want.
- **Expect exemptions.** Personal privacy, ongoing investigation, deliberative process, trade secret, and security exemptions are normal. The redaction pattern itself is often informative.
- **Track the timeline.** Statutory response deadlines exist and are frequently missed; a polite follow-up citing the statute is normal practice.
- **Understand the record.** A request is itself a public act in many jurisdictions, meaning your interest may become visible.

1.6.5 The Coverage Trap, Illustrated

An analyst searches a state's business registry for a company and finds nothing. Three wrong conclusions and one right one:

Wrong: the company does not exist. Wrong: the company is fraudulent. Wrong: the company operates informally.

Right: the company is not registered *in that state under that exact name*. It may be registered elsewhere, under a former name, as a differently structured entity, as a foreign entity qualified to do business, or under a trade name filed at the county rather than the state. **The absence of a record is a statement about the record system, not about the world.** Every negative finding in this category must be written with its coverage boundary attached.

1.7 Court Records

1.7.1 Why They Are Worth Their Own Section

Court records are government records with three special properties that justify separate treatment.

They are adversarial. Two parties with opposing interests each put their strongest version on the record, and each attacks the other's. The result is a body of contested claims tested by opposing

counsel, which is a materially different evidentiary object from a one-sided filing.

They contain compelled disclosure. Discovery produces documents the parties never intended to publish. Exhibits attached to filings are frequently internal material: emails, contracts, financial records, technical reports. Some of the most consequential public revelations about companies have come from exhibits in ordinary civil litigation.

They are indexed by party name. This makes them one of the few categories where you can systematically ask "everything involving this entity" and get a bounded, checkable answer.

1.7.2 The Structure of a Case File

Understanding the anatomy prevents the most common beginner error, which is misreading an allegation as a finding.

CASE	
—	DOCKET (the index) Numbered list of every filing, with dates. Read this first. Tells you what exists and the sequence of the case.
—	COMPLAINT / PETITION ← ALLEGATIONS ONLY One party's claims. Not evidence. Not findings. Frequently quoted in the press as though it were fact.
—	ANSWER / RESPONSE The other party's denials and defenses.
—	MOTIONS AND BRIEFS Legal arguments, often with factual assertions.
—	EXHIBITS ← OFTEN THE MOST VALUABLE MATERIAL Contracts, emails, financials, reports, photographs. Attached to filings and therefore public unless sealed.
—	DEPOSITIONS AND TESTIMONY (when filed) Sworn statements. High evidentiary weight.
—	ORDERS AND OPINIONS ← FINDINGS BY THE COURT What a judge actually decided and why. The authoritative content.
—	JUDGMENT / DISPOSITION The outcome: verdict, settlement, dismissal, default. Note that a dismissal or settlement is not a finding of fact.

The single most important reading rule in this section: a complaint is an allegation, an order is a finding, and a settlement resolves a dispute without establishing facts. An analyst who writes "court records show the company falsified reports" when the source is a complaint has made a serious error with defamation exposure attached.

1.7.3 Court Systems and Their Access Models

System type	What it holds	Access
Federal trial and appellate courts	Civil, criminal, bankruptcy,	Electronic case management

System type	What it holds	Access
	appeals	with per-page fees; free opinion access; free third-party mirrors of many documents
State trial courts	The large majority of all litigation, including most contract, tort, family, probate, and landlord-tenant matters	Extremely variable: some statewide free portals, some county-by-county, some terminal-only at the courthouse
State appellate and supreme courts	Opinions and briefs	Usually free online
Administrative tribunals	Labor, immigration, tax, utility, workers compensation, professional discipline	Variable; often overlooked and often rich
Bankruptcy courts	Creditor lists, asset schedules, insider transactions	Same system as federal courts; enormously informative about a company's real financial condition
International and regional courts	Treaty and human rights matters	Usually published

The practical coverage warning. Most litigation happens in state trial courts, and state trial court access is the worst in the system. A federal docket search that returns nothing has told you almost nothing about whether an entity has been sued. Any litigation finding must state which systems were searched.

1.7.4 The Six Properties

Property	Assessment
Content	Pleadings, motions, exhibits, testimony, orders, judgments, liens, bankruptcies, criminal dispositions
Production process	Filed under rules of court, subject to adversarial testing and judicial review. Very strong process for orders; no process at all for the truth of allegations
Access model	Free to fee, portal to in-person, wildly variable by jurisdiction
Coverage	Excellent where digitized; state trial courts are the weak point. Sealed and expunged matters are invisible by design
Latency	Days for filings, months to years for dispositions
Failure mode	Reading allegations as findings; assuming a name match is an identity match; missing cases filed

Property	Assessment
	under a subsidiary or former name

1.7.5 Name Matching and Identity

Two people share a name. Companies use similar names deliberately. The discipline: never assert identity from a name match alone. Corroborate with at least one additional identifier that appears in the record: address, date range, counsel of record, related case numbers, entity registration number, or a subject matter that matches. In products, write "a party of the same name" until you have that second identifier, then say what the identifier was.

1.8 Radio

1.8.1 The Category With the Longest Intelligence Pedigree

Chapter 2's history section began the modern discipline with wartime broadcast monitoring. Radio remains analytically live for four reasons, three of which have nothing to do with nostalgia.

It is where information goes when infrastructure fails. In a disaster with power and network outages, local radio may be the only functioning information channel, both for the affected population and for anyone monitoring from outside.

It carries local content that is never written down. Small-market local radio covers municipal meetings, school closures, agricultural conditions, and community disputes that no digital outlet records.

It is a primary channel in low-connectivity regions. For large parts of the world, radio remains the dominant mass medium, and monitoring it is the only way to know what the local information environment contains.

Public safety and utility radio is partially monitorable. Where lawful in the jurisdiction, unencrypted public safety transmissions are receivable, and aggregated streaming of scanner audio exists. Legality varies substantially by country and by state, and increasing encryption has reduced availability. Treat this as jurisdiction-dependent and confirm before relying on it.

1.8.2 The Six Properties

Property	Assessment
Content	News, talk, local information, emergency broadcasts, official statements, shortwave international broadcasting, amateur and public safety transmissions
Production process	Licensed broadcasters with editorial practice; talk programming with none; official broadcasts as

Property	Assessment
	primary statements of a government's public position
Access model	Live listening, station streaming, aggregated archive services, monitoring services, and library or institutional broadcast archives
Coverage	Very strong locally and in low-connectivity regions; weak archival preservation, since much broadcast content is never recorded or retained
Latency	Real time
Failure mode	Ephemerality. If you did not record it, it is gone. Also, state broadcasting is a statement of official position, not a report of fact

1.8.3 Working With Radio Analytically

For monitoring, capture. Recording as you listen is the difference between a citable finding and a memory. For historical work, broadcast archives at national libraries and university collections hold material that exists nowhere else, and monitoring service transcripts from government open source programs have been partially released and digitized. For state broadcasting, analyze it the way early Cold War analysts did: not as a source of facts but as evidence of what a government wants believed, with changes in emphasis, terminology, and personnel mentions as indicators.

1.9 Television

1.9.1 What Television Uniquely Contributes

Footage. Television produces and archives visual records of events, locations, and people. For geolocation, chronolocation, and physical reconstruction, broadcast footage is frequently the highest-quality visual material available for events before the smartphone era.

Local news archives. Local television covered crimes, fires, meetings, and openings in visual detail, and station archives and library collections hold the tape.

On-record statements. An official's televised statement is a primary source for what they said, when, and in what words. Transcripts and closed-caption records make this searchable.

Live continuous coverage. During a developing event, continuous broadcast coverage produces a timestamped visual record that is often more precise than later written accounts.

1.9.2 The Six Properties

Property	Assessment
Content	News footage, interviews, documentaries, official statements, live event coverage, weather and traffic imagery
Production process	Editorial selection with strong visual verification incentives; framing and time constraints introduce compression
Access model	Broadcaster archives, library and university television news archives, closed-caption transcript databases, clip services, and platform-posted segments
Coverage	Strong for major markets and major events; thin for small markets; preservation is inconsistent, and some archives are being lost
Latency	Real time to hours
Failure mode	Framing. What is out of frame is invisible, and a camera position is an editorial choice. Also, clips circulate stripped of context and are routinely reused misleadingly

1.9.3 Analytic Technique With Broadcast Footage

Treat footage the way Chapter 2 taught you to treat any visual material: establish provenance (which broadcaster, which date, which segment), geolocate against fixed features, chronolocate where possible, and check for reuse. Closed-caption archives are the best entry point for finding footage, because captions are text and therefore searchable, which lets you locate the exact broadcast and timestamp where a subject was discussed.

1.10 Podcasts

1.10.1 Why a Chapter on Sources Includes Podcasts

Two reasons, and the first surprises people.

People say things on podcasts that they would never put in writing. The format is long, conversational, and feels informal, and interviewees relax. Executives describe strategy, timelines, and internal disagreement. Technical staff describe architecture and tooling. Founders describe customer concentration. This is voluntarily published material that is often more specific than anything in the company's own communications, and it is a well-established open source in competitive and cybersecurity research.

Industry and niche podcasts cover subjects with no other coverage. Trade podcasts discuss specific

facilities, regulations, equipment, and market conditions in detail that no article matches.

1.10.2 The Six Properties

Property	Assessment
Content	Long-form interviews, industry discussion, technical deep dives, true crime and investigative series, official and corporate podcasts
Production process	Almost none. Minimal editing, no fact-checking, no correction practice at most shows. Statements are unvetted
Access model	Free, open RSS distribution, platform hosting, and increasingly machine-generated transcripts
Coverage	Very strong for technology, business, and niche industry; weak for anything before roughly 2010
Latency	Days to weeks from recording to release
Failure mode	Treating an offhand conversational statement as a considered corporate position. People misspeak, round numbers, and exaggerate in conversation

1.10.3 Working With Audio at Scale

The practical problem is that audio is unstructured and slow to consume. The professional approach: locate transcripts where they exist, since many platforms and shows now publish them; search transcript text rather than listening; when you find a relevant passage, listen to it in context before quoting it, because transcripts contain errors and conversational statements are easy to misread when stripped of tone and qualification; and always cite the episode, the date, and the timestamp so someone else can check you.

A calibration note. A statement made conversationally deserves a lower credibility rating than the same statement in a filing. Rate it accordingly and say why. "The CEO stated in a podcast interview" is honest framing; "the company disclosed" is not.

1.11 Satellite Imagery

1.11.1 What It Establishes That Nothing Else Does

Physical ground truth, and change over time. Imagery answers questions of the form: does this facility exist, is it operating, how large is it, what is stored there, has it expanded, was it damaged, how many vehicles are present, is the field planted, is the reservoir full.

Chapter 2 emphasized that imagery converts an argument into an exhibit. That is its central analytic property: a photograph of a construction site is checkable by anyone, which makes it unusually

resistant to dispute.

1.11.2 Types of Imagery and What Each Is For

Type	What it captures	Best used for	Limits
Optical (visible light)	What a camera would see	General assessment, counting, identification	Requires daylight and clear skies
Multispectral	Additional bands beyond visible	Vegetation health, water, soil, burn scars, material differences	Requires interpretation training
Thermal / infrared	Heat signatures	Whether a facility is operating, fire detection, night activity	Coarse resolution typically
Radar (SAR)	Radar returns	Sees through clouds and at night; detects surface change and vessels	Non-intuitive appearance; needs training to read
Aerial and drone	Lower-altitude imagery	Much higher resolution; oblique angles	Limited coverage; drone use is heavily regulated
Street-level	Ground-level photography of roads and frontages	Geolocation reference, facility frontage, signage, historical comparison	Only where roads exist; capture dates are sporadic

1.11.3 The Six Properties

Property	Assessment
Content	Physical state of the earth's surface at specific times
Production process	Instrument capture. No human intent in the record, which is an unusual reliability advantage; the interpretation is where error enters
Access model	Free moderate-resolution government archives, free consumer platforms with mixed dates, commercial high-resolution by purchase or subscription, and free access programs for humanitarian and journalistic use
Coverage	Global, but with variable revisit rates, cloud obstruction, and inconsistent archive depth. Consumer platform imagery may be years old and its date is not always displayed
Latency	Same-day for tasked commercial collection; days to weeks for free archives; years for consumer platform base layers

Property	Assessment
Failure mode	Assuming the image is current. Also over-reading: distinguishing a facility's type or function from overhead requires training, and confident misinterpretation of imagery has caused real historical errors

1.11.4 The Discipline of Change Detection

The single most valuable technique in this category, and it costs nothing but care.

1. **Fix the location precisely.** Coordinates, not a place name.
2. **Establish the capture date of every image you use.** If a platform does not show it, find one that does, or treat the date as unknown and say so.
3. **Compare the same frame across dates.** Same coordinates, same zoom, same orientation. Differences in framing create false apparent change.
4. **Describe what changed in observable terms.** "A structure measuring approximately 60 by 30 metres appears between the March and August images" is observation. "They built a warehouse" is interpretation, and the interpretation goes in a separate sentence with a confidence level.
5. **Corroborate the interpretation from another category.** A building permit, a procurement award, a job posting, or a news item. Imagery plus documents is the strongest common pairing in open source work.

1.11.5 Reading Imagery Responsibly

Three cautions worth internalizing. Resolution limits what can be claimed: at moderate resolution you can see that vehicles are present, not what kind. Shadows are your friend for estimating height and time of day and your enemy for identifying objects. And scale errors are the most common beginner mistake; use a known reference (a road width, a standard shipping container, a rail car) before asserting any dimension.

1.12 Social Media

1.12.1 The Category With the Best Timeliness and the Worst Process

Social media is simultaneously the fastest source available and the least reliable, and the reason is the same for both: there is no production process at all. No obligation, no schedule, no editorial review, no correction practice, no penalty for falsehood.

That does not make it low value. It makes it a category where the analytic burden sits almost entirely on verification rather than on collection.

1.12.2 What It Actually Provides

Contribution	Explanation
Eyewitness material	Participant photographs and video from inside an event, often minutes after it
Geolocatable visual content	Imagery containing fixed features that can be located and dated
Self-disclosure by organizations	Corporate accounts announce openings, hires, partnerships, and products
Self-disclosure by individuals in professional capacity	Employment history, technology stack, conference attendance, professional affiliations
Sentiment and narrative	What is being said about a subject, and by whom, and how it is spreading
Network structure	Who is connected to whom, at the organizational level
Recruitment and vendor signals	Job posts, vendor endorsements, technology mentions

1.12.3 The Six Properties

Property	Assessment
Content	Text, images, video, audio, live streams, comments, profile data, engagement metrics
Production process	None
Access model	Public viewing where accounts are public; programmatic access heavily restricted in recent years; terms of service constrain automated collection
Coverage	Biased toward connected, younger, urban, and platform-specific populations. Platform mix varies enormously by country and demographic. Deleted content vanishes
Latency	Seconds
Failure mode	Everything Chapter 2 warned about: recycled content, coordinated inauthentic behavior, synthetic media, circular amplification, and misidentification of individuals

1.12.4 The Verification Sequence, Restated for This Category

Because this is where errors concentrate, the sequence from Chapter 2 is repeated here as a checklist to

apply to every social media item you intend to use:

1. Reverse image or video search for earlier appearances.
2. Identify the earliest publisher and timestamp, not the account you saw it from.
3. Assess whether that publisher plausibly had access to what they published.
4. Geolocate from fixed features if the content is visual.
5. Chronolocate from shadows, weather, or visible dated objects.
6. Look for an independent second publisher of the same moment from a different angle.
7. Examine account history for age, topic consistency, and network.
8. Check for phrasing repetition across accounts claiming independence.
9. Archive the item, because it may be deleted.
10. Rate reliability and credibility separately, and state what you could not establish.

1.12.5 The Ethical Constraint, Specifically

This is the category where Rule 1 is most easily violated, because it is trivially easy to compile a detailed profile of a private person from public posts. Restate the standard: individuals appear in your work only in a professional capacity relevant to a stated requirement, never as subjects in themselves, never as family members or associates of a subject, and never if they are minors. The mosaic effect from Chapter 2 is not a theoretical concern in this category; it is the default outcome of unstructured browsing.

1.13 Maps

1.13.1 Maps as Analytic Instruments

A map is not a picture of a place; it is a selective model of a place, made for a purpose. Understanding the purpose tells you what the map includes and, more importantly, what it omits.

The investigative uses:

Establishing spatial relationships. What is adjacent to what, how far apart, which route connects them, what is upstream, what is in the floodplain.

Geolocation reference. Matching visual features in an image against mapped features.

Historical comparison. Historical maps establish what existed, when, under what name, and with what boundaries. Property disputes, environmental legacy questions, and urban history all turn on this.

Jurisdictional determination. Which city, county, district, or authority governs a location, which determines which records exist and where to request them. This is the most underrated use of maps in open source work, because it directly drives collection planning.

1.13.2 Types of Maps and Data

Type	Content	Investigative use
Topographic	Terrain, elevation, watercourses, structures	Terrain analysis, line of sight, drainage
Cadastral / parcel	Property boundaries and ownership	Ownership, tenancy, parcel identification
Zoning and land use	Permitted uses by area	Whether a claimed use is even permissible at an address
Hazard and floodplain	Flood, seismic, wildfire, subsidence exposure	Risk assessment, disaster planning
Infrastructure and utility	Roads, rail, pipelines, power, water	Dependency and disruption analysis
Administrative boundary	Jurisdictions, districts, service territories	Determining which records system to query
Nautical and aeronautical charts	Depths, hazards, airspace, approaches	Maritime and aviation analysis
Historical	Prior conditions, former names, former boundaries	Change over time, legacy land use
Collaborative / open	Community-contributed geographic data	Detailed local features, rapid disaster mapping

1.13.3 The Six Properties

Property	Assessment
Content	Geographic features, boundaries, ownership, hazard exposure, infrastructure
Production process	Government mapping agencies with survey standards, commercial cartography, and community contribution with variable verification
Access model	Free government geospatial portals, county parcel viewers, free open mapping projects, commercial platforms, and library historical map collections
Coverage	Excellent in wealthy jurisdictions; sparse in others. Community-mapped areas vary from exhaustive to blank
Latency	Parcel and zoning data updates on cycles of weeks to years; base maps update irregularly
Failure mode	Treating a map as current, and treating a boundary as authoritative when the definitive record is a legal description rather than a rendering

1.13.4 The Jurisdiction Workflow

A practical routine worth memorizing, because it unlocks the government records category:

1. Take the address or coordinates.
2. Determine the county or equivalent, the municipality if any, and any special districts (water, fire, school, utility).
3. For each of those, identify the records systems: assessor, recorder, permits, code enforcement, planning, and the relevant court.
4. Now you know exactly which portals to query, and you can state your coverage boundary precisely in the product.

Most students who cannot find a record are searching the wrong jurisdiction's system.

1.14 Corporate Filings

1.14.1 Compelled Disclosure Is the Whole Point

Corporate filings are valuable for one structural reason: companies file them because the law requires it, in a defined format, on a schedule, with penalties for material misstatement. A company's marketing describes what it wants believed. Its filings describe what it is willing to certify.

1.14.2 The Filing Landscape

Filing type	What it contains	Where
Formation and registration	Entity name, type, formation date, registered agent, officers or managers, status	State or national corporate registry
Annual or biennial reports	Updated officers, addresses, and status	Same registry
Trade name / DBA filings	Operating names distinct from the legal name	Often county-level, not state
Foreign qualification	Registration to do business in a state other than the state of formation	Each state where qualified
Public company periodic reports	Audited financials, business description, risk factors, legal proceedings, management discussion, executive compensation, related-party transactions	Securities regulator's public filing system

Filing type	What it contains	Where
Public company event reports	Material events: leadership changes, acquisitions, defaults, auditor changes, restatements	Same
Ownership and insider reports	Who holds significant stakes; insider transactions	Same
Prospectuses and offering documents	Detailed business and risk description ahead of raising capital	Same
Beneficial ownership registries	Natural persons behind entities, where such registries exist	National registries, variable by country and subject to legal change
UCC filings	Security interests in personal property; identifies lenders and encumbered assets	State filing office
Liens and judgments	Tax liens, mechanics liens, judgment liens	County recorder and courts
Nonprofit information returns	Revenue, expenses, officers, compensation, grants, related organizations	National tax authority public disclosure

1.14.3 The Sections That Repay Careful Reading

For a public company, four sections routinely contain material nobody reads:

Risk factors. Written by counsel to disclose what could go wrong. Read as a list of what management actually worries about. Changes between years are the signal: a newly added risk factor means something changed.

Legal proceedings. Companies must disclose material litigation. Compare this against what you find in dockets; discrepancies are findings.

Related-party transactions. Business done with insiders and their affiliates. Frequently the most interesting page in the document.

Management discussion and analysis. Management's own explanation of results, including customer concentration, supplier dependency, and segment performance. Concentration disclosures answer the question that Chapter 2's Lab 4 could not.

A note on nonprofit returns. For any nonprofit, the annual information return is remarkably informative: officer compensation, largest vendors, grants made and received, related organizations, and governance practices. Because it is a tax filing, it is made under penalty and is fully public.

1.14.4 The Six Properties

Property	Assessment
Content	Entity identity, structure, ownership, financials, obligations, litigation, insiders
Production process	Legal obligation, defined forms, audit for public company financials, penalties for material misstatement. Among the strongest processes of any category
Access model	Mostly free national systems for public companies; state registries free to modest fee; some jurisdictions charge per document
Coverage	Excellent for public companies and registered entities. Very thin for private companies in jurisdictions with minimal disclosure requirements, which is most of them
Latency	Days for event reports, weeks to months for periodic reports, annual for registry reports
Failure mode	Assuming a private company's absence of filings means absence of activity; and reading a company's own characterization as fact rather than as a certified assertion

1.15 Financial Records

1.15.1 What Is Public and What Is Not

The boundary here trips people up, so state it plainly. **Individual bank accounts, transactions, and credit histories are not open sources.** They are legally protected, and accessing them without authorization is a crime. What is publicly available is a different set of things:

Publicly available	Not publicly available
Public company audited financial statements	Private company financials, absent a filing obligation
Nonprofit information returns	Individual bank and brokerage records
Municipal bond offering documents and continuing disclosure	Individual credit reports (regulated access only)
Government budgets, audits, and spending records	Individual tax returns
Contract awards and payment records	Payment card transaction data
Bankruptcy schedules of assets and liabilities	Internal management accounts

Publicly available	Not publicly available
Property assessments and tax rolls	Payroll records
Liens, UCC filings, and judgments	
Regulatory financial filings (banks, insurers, utilities)	
Sanctions and debarment lists	
Trade and customs statistics, and in some jurisdictions shipment-level import records	

1.15.2 Government Financial Records Are the Underused Half

Public sector financial transparency is extensive and almost nobody uses it. Municipal budgets and audited financial statements disclose fund balances, debt service, pension obligations, and capital plans. Bond offering documents are prospectus-quality descriptions of a jurisdiction's finances, infrastructure, litigation, and risks, and continuing disclosure obligations mean they are updated. Contract and grant award databases show who is paid what for what. Check registers, published by many jurisdictions, list individual payments to vendors.

For any investigation involving a public agency, a vendor to a public agency, or a community's condition, these are primary, structured, compelled records.

1.15.3 The Six Properties

Property	Assessment
Content	Statements, budgets, spending, obligations, liens, assessments, trade flows
Production process	Audit, regulatory filing, statutory publication. Strong where present
Access model	Free public systems for most government and public company material; some commercial data is fee-based
Coverage	Strong for public entities and public companies; sparse for private entities and individuals, by design
Latency	Quarterly and annual cycles; months for audits
Failure mode	Comparing figures across entities without checking accounting definitions, fiscal year boundaries, or currency; and over-interpreting a single period

1.15.4 Reading Financial Material Without an Accounting Background

Four things a non-specialist can do reliably:

1. **Compare across time rather than across entities.** Trend within one entity is far safer than comparison between two, because definitions differ.
 2. **Read the notes.** The notes to financial statements contain the contingencies, commitments, related parties, and accounting policy choices. The statements without the notes are misleading by omission.
 3. **Look for what changed.** A new auditor, a restatement, a changed accounting policy, a newly disclosed contingency: each is a finding.
 4. **Say what you do not know.** A non-specialist can report what a document discloses without asserting a conclusion about financial health. "The filing discloses X" is honest; "the company is in financial distress" requires expertise you may not have and should be flagged as a gap.
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1.16 Public Databases

1.16.1 Structured Query Instead of Keyword Search

A public database is a structured, queryable collection maintained by an authority. This is the category that most rewards learning the system rather than searching for text, because the answer is definitive and bounded rather than probabilistic.

1.16.2 A Working Inventory

Domain	Database types
Corporate and legal	Corporate registries, UCC indexes, court case indexes, bankruptcy indexes, patents, trademarks
Licensing and discipline	Professional licensing boards, contractor licensing, healthcare provider registries, disciplinary actions
Safety and enforcement	Product recalls, vehicle safety complaints and investigations, workplace safety inspections and citations, food safety inspections, transportation accident records
Environmental	Emissions and release inventories, permits, enforcement actions, contaminated site inventories, water quality monitoring
Transportation	Aircraft and vessel registries, aircraft position feeds, vessel position feeds, carrier safety records, rail incident records
Government operations	Contract and grant awards, lobbying and

Domain	Database types
	campaign finance disclosures, sanctions and debarment lists, budget and spending portals, employee salary rosters
Statistical	Census and demographic products, labor statistics, agricultural statistics, trade statistics, health surveillance
Scientific and technical	Genomic and biological repositories, clinical trial registries, standards registries, spectrum licenses
Internet infrastructure	Domain registration data, certificate transparency logs, autonomous system and address allocation registries, passive DNS collections, internet-wide scan data

1.16.3 The Six Properties

Property	Assessment
Content	Structured records within a defined scope
Production process	Mandated reporting into a defined schema, with agency validation of varying rigor
Access model	Free web query, bulk download, sometimes programmatic access, sometimes fee
Coverage	Definitive within scope and jurisdiction; silent outside it. Historical depth varies by system
Latency	Real-time for some feeds; quarterly or annual for statistical products
Failure mode	Misreading a field. Every database has a schema and a codebook, and fields mean what the codebook says they mean, not what the label suggests

1.16.4 The Codebook Rule

The most important habit in this category: **before you use a field, read its definition.** Databases are full of fields whose plain-language names mislead. "Inspection date" may be the date of the report rather than the visit. "Employees" may be a self-reported estimate at registration and never updated. "Status: active" may mean only that the entity has not been administratively dissolved, not that it operates. A finding built on a misread field is confidently wrong, which is the worst kind of wrong.

Every serious public database publishes documentation: a codebook, a data dictionary, a technical guide, or field-level help. Reading it takes ten minutes and prevents the majority of database errors.

1.17 Archived Websites

1.17.1 Why This Is a Category and Not a Trick

Websites are edited without notice. Pages are removed. Claims are quietly revised. An organization's site today is a statement of what it currently wants to say, and it carries no record of what it said before.

Web archives change that. They convert a mutable surface into a dated, citable sequence, which makes them one of the highest-value categories in this chapter and the one most often overlooked by beginners.

1.17.2 What Archives Are For

Use	Explanation
Establishing what a party claimed at a specific time	The claim as of a date, not as of today
Detecting quiet revision	Removed claims, changed numbers, deleted pages, altered terms
Recovering deleted content	Pages, documents, and profiles that no longer exist live
Dating a change	Bracketing when something appeared or disappeared, to the precision of capture intervals
Reconstructing an organization's history	Leadership, locations, clients, products, and positioning over years
Recovering linked documents	Archives frequently hold PDFs and files no longer on the live site
Detecting evidence of preparation	A page created before its public announcement

Chapter 2's Lab 4 turned on exactly this: an archived page showing that a four-facility claim postdated a groundbreaking announcement.

1.17.3 The Archive Landscape

Type	Description	Notes
Large general web archives	Broad, long-running crawls of the public web with dated captures	The primary resource; coverage varies enormously by site popularity
National library web archives	Legal-deposit collections of a country's web presence	Often better national coverage than general archives; access sometimes restricted to reading rooms
Topical and event archives	Curated collections around events, elections, or subjects	Deeper coverage of the covered topic

Type	Description	Notes
Search engine caches	Very recent snapshots	Short-lived and increasingly unavailable; not a substitute for an archive
Your own archiving	Captures you make at the moment of collection	The only capture you control, and therefore essential
Archive-on-demand services	Let you request a capture of a live page now	Use these when you find something important; they timestamp the state you saw

1.17.4 The Six Properties

Property	Assessment
Content	Prior states of web pages and linked files
Production process	Automated crawling, on schedules driven by site popularity and crawl policy. No human curation of individual captures
Access model	Mostly free
Coverage	Deeply uneven. Popular sites captured frequently, obscure sites rarely or never. Sites can request exclusion. Dynamic, script-driven, and login-gated content often fails to archive. Images and files may be missing from a capture
Latency	Captures exist from days to years apart depending on the site
Failure mode	Treating capture intervals as precise dating, and assuming that content absent from a capture was absent from the site

1.17.5 Method for Comparing Captures

The technique Lab 4 will have you perform:

1. **List available captures** for the URL and record the dates. The pattern of capture frequency is itself informative.
2. **Select comparison points** with a purpose: before and after a known event, annually for a long trend, or bracketing a suspected change.
3. **Compare systematically, not impressionistically.** Work through defined elements: page title, navigation menu items, headline claims, numeric claims, named clients or partners, leadership names and titles, locations, terms and policies, contact details, and linked documents.

4. **Record changes in a table** with the capture dates that bracket each change, and state the precision honestly: "changed between 12 March and 4 July" not "changed in May."
 5. **Distinguish substantive from cosmetic change.** A redesign is not a finding. A removed client name is.
 6. **Check for what a capture missed.** If images or a linked PDF failed to capture, note it rather than concluding the content was not there.
 7. **Cite the capture URL**, which contains the timestamp, so a reader can verify.
-

1.18 The Dark Web

1.18.1 First, Fix the Terminology

The three terms are constantly conflated, including by professionals and by journalists. Get this right, because the distinction has practical consequences.

Term	Definition	Example	Reachable how
Surface web	Crawlable, indexed pages	A company's public homepage	Any browser, findable by search
Deep web	Content on the ordinary internet that is not indexed: behind queries, forms, logins, or paywalls, or excluded from crawling	A court case management system, a library catalog, your webmail, an academic database	Any browser, but you must go there and query it
Dark web	Content requiring specific overlay network software to reach, using addresses that do not resolve in ordinary DNS	Sites on anonymizing overlay networks	Specialized client software

The correction that matters most: the deep web is not sinister. It is where nearly all of the authoritative material in this chapter lives. Court records, filings, databases, and library holdings are deep web. Anyone who uses "deep web" and "dark web" interchangeably is signaling that they have not understood the landscape.

1.18.2 What the Dark Web Actually Contains

A realistic inventory, because both the popular exaggeration and the dismissive minimization are wrong:

- Criminal marketplaces (narcotics, stolen credentials, fraud services, malware).
- Data leak and extortion sites operated by ransomware groups, publishing victim names and

stolen documents.

- Forums, including criminal forums and legitimate privacy and security communities.
- Mirrors of legitimate services: news organizations, search engines, email providers, and secure document submission systems for journalists and whistleblowers.
- Content in and about jurisdictions where ordinary publication carries personal risk, including human rights reporting and censorship-circumvention resources.

The last two categories are why the network exists and why sweeping characterizations of it are wrong. It is infrastructure, and infrastructure is used by whoever finds it useful.

1.18.3 Why It Matters Analytically

For cybersecurity work in particular, dark web monitoring supports:

- **Breach discovery.** An organization frequently learns it was breached because its data appears on a leak site. Ransomware groups publish victim lists as an extortion tactic, which makes those sites an early indicator.
- **Credential exposure.** Stolen credentials for an organization's users appearing for sale.
- **Threat actor tracking.** Advertisements, recruitment, and technique discussion that inform defensive priorities.
- **Third-party risk.** A vendor's appearance on a leak site is your exposure too.
- **Fraud and counterfeit.** Discussion of methods targeting a specific brand or sector.

1.18.4 Why You Will Not Do This in This Course

Rule 4 from the front of this chapter, with its reasoning:

Legal exposure. Merely viewing some content on these networks is a criminal offense in many jurisdictions, and the content is not labeled in advance. Encountering child sexual abuse material is a realistic risk on unfiltered exploration, and there is no lawful "I was just looking" position.

Operational risk. These environments are actively hostile: malware distribution, deanonymization attempts, and law enforcement operations. Doing this from a personal or institutional device is a serious mistake.

Psychological risk. Analysts who work these environments professionally require support structures because the content is genuinely harmful to encounter.

It is unnecessary for learning. Everything a student needs to learn about this category is conceptual: what it is, what it holds, why it matters, and how professionals handle it.

How professionals actually handle it. Not by browsing. Through commercial monitoring services that index leak sites and marketplaces and deliver alerts; through sector information-sharing organizations; through law enforcement liaison; and, where an organization does conduct direct collection, under written legal authorization, on isolated infrastructure, by trained staff, with defined scope, logging, and

a documented handling procedure for illegal content. If you ever work in this area, that is the framework you should insist on before touching anything.

1.18.5 The Six Properties

Property	Assessment
Content	Criminal markets, leak and extortion sites, forums, legitimate privacy-preserving mirrors and services
Production process	None, and actively adversarial. Fraud among criminals is routine, so claims on these sites are unusually unreliable
Access model	Overlay network client software; monitoring services for indirect access
Coverage	Unmappable; sites appear and vanish; no index is complete
Latency	Real time, but discovery is unreliable
Failure mode	Believing claims made by criminals. Also, false victim listings on leak sites do occur, and a listing is an allegation, not proof of a breach

1.19 Metadata

1.19.1 What Metadata Is

Metadata is data about data: information a file or system records about content, separately from the content itself. It is generated automatically by devices and software, which is precisely why it is analytically valuable. Content is authored deliberately; metadata is usually produced without anyone thinking about it.

1.19.2 Where Metadata Lives

Container	Typical fields
Photographs (EXIF, IPTC, XMP)	Camera make and model, lens, exposure settings, orientation, original capture timestamp, modification timestamp, software used, GPS coordinates and altitude where recorded, thumbnail of the original image, sometimes a serial number or owner name
PDF documents	Title, author, subject, keywords, creating application, PDF producer, creation timestamp, modification timestamp, and in many cases embedded fonts, layer structure, form fields, and

Container	Typical fields
	remnants of prior edits
Office documents (word processing, spreadsheets, presentations)	Author, last modified by, company, total editing time, revision number, creation and modification timestamps, template used, comments, tracked changes, hidden rows or slides, embedded objects and their own metadata
Audio and video	Duration, codec, bitrate, device, creation timestamp, sometimes GPS, sometimes software history
Email	Full header chain: originating server and address, relay path with timestamps, message ID, client software, authentication results, reply-to and return-path discrepancies
Web pages	Generator tags, analytics identifiers, comment blocks, server headers, structured data markup, file paths in asset URLs
Filesystem	Created, modified, accessed timestamps, owner, permissions, and file size
Archive files	Per-entry timestamps, original path structure, and sometimes original usernames in paths

1.19.3 What Metadata Can Establish

Question	Metadata that speaks to it
When was this created?	Creation timestamp, capture timestamp, filesystem dates, email relay timestamps
Was it modified after creation?	Modification timestamp later than creation, revision number, producer chain
Where was it captured?	GPS coordinates in image or video metadata
What device made it?	Camera make and model, serial number, device identifier
Who authored it?	Author and last-modified-by fields, company field, email address in headers
What software was used?	Creating application and producer strings, generator tags
Are two files related?	Same device, same author, same template, same producer version, same analytics identifier across sites
How long was it worked on?	Total editing time and revision count in office documents

Question	Metadata that speaks to it
Is the claimed origin plausible?	Whether the device, timestamps, and software are internally consistent with the claim

1.19.4 Worked Illustration

Consider a document circulated as a leaked internal memo, allegedly written in March by a company's operations director.

Metadata field	Value	What it suggests
Creating application	A consumer word processor, not the enterprise suite the company uses in its other public documents	Inconsistent with the claimed origin
Author	A generic default name	Uninformative, but notable that it is not the claimed author
Creation timestamp	Two days before the document surfaced, not March	Directly contradicts the claimed date
Modification timestamp	Six minutes after creation	Very short authoring time for a substantive memo
Total editing time	4 minutes	Consistent with transcription or paste, not drafting
Template	None	Inconsistent with a company that uses letterhead templates elsewhere

None of these fields proves fabrication. Metadata can be edited, stripped, or regenerated by intermediate handling, and a genuine document can be re-saved by a recipient, which resets several fields. But the **convergence** here is strong, and combined with the absence of any corroborating source it supports an assessment that the document's claimed provenance is not supported, at moderate to high confidence. That is exactly the reasoning pattern from Chapter 2: convergence of independent weak indicators.

1.19.5 The Limits, Stated Carefully

This subsection is the one most likely to appear on an exam, because the limits are what separate careful from careless use.

Metadata is corroborating, not conclusive. Every field can be edited with ordinary tools. A timestamp is an assertion by a device or an application, not an observation by a neutral witness.

Platforms strip it. Most social platforms remove image metadata on upload, including GPS. The absence of GPS in a downloaded social media image tells you nothing about the original.

Handling changes it. Emailing, re-saving, converting, or screenshotting a file rewrites metadata. A modification timestamp may reflect a recipient's action, not the author's.

Device clocks are wrong. Cameras with unset clocks produce timestamps years off. Time zones are frequently unrecorded or misrecorded.

Absence is ambiguous. Missing metadata may mean stripping, conversion, deliberate removal, or a device that never recorded it.

GPS can be inaccurate. Coordinates may be derived from network positioning rather than satellite fix, with error measured in hundreds of metres or worse.

The correct analytic posture: use metadata to generate and test hypotheses, corroborate it against content and against other categories, state which fields you relied on, and never rest a judgment on a single field.

1.19.6 The Defensive Side

Metadata is also an exposure. Organizations routinely publish documents containing author names, internal file paths revealing directory structures and usernames, tracked changes revealing what was removed, comments never intended for publication, and embedded original images beneath cropped versions. Photographs published by individuals have revealed home locations through embedded GPS.

The practical control is a publication process that strips metadata from documents and images before release, and a periodic audit of what an organization's already-published files disclose. Lab 5 will have you perform exactly that audit on sample files.

1.20 Choosing Among Categories

1.20.1 The Selection Heuristic

Before searching, ask what kind of thing the answer is. The answer type points at the category.

If the answer is...	Start with
A legal fact (ownership, obligation, judgment, status)	Court records, corporate filings, government records
A physical fact (existence, size, condition, change)	Satellite imagery, maps, permits
A financial fact	Corporate filings, financial records, government spending data
A statement someone made	Television, radio, podcasts, social media, filings, testimony
A historical fact	Newspaper archives, books, archived websites, historical maps
A general rate, method, or mechanism	Academic papers, technical standards,

If the answer is...	Start with
	government statistics
A current event fact	News, social media, official statements, then corroborate elsewhere
A technical or infrastructure fact	Public databases (domain, certificate, scan, registry data)
A prior claim now removed	Archived websites
A provenance or authenticity question	Metadata, plus reverse image search
A criminal-ecosystem fact	Dark web monitoring services, not direct access

1.20.2 The Corroboration Rule

Corroboration across categories is worth far more than corroboration within a category. Three news articles agreeing is weak, because they may share an origin and they share a production process and its biases. A news article, a permit record, and a satellite image agreeing is strong, because the three could not be coordinated cheaply and their failure modes are unrelated.

Build this into your product structure: for each significant finding, state which categories support it. A finding supported by one category, however many items within it, should be labeled as such.

1.20.3 A Worked Selection Example

Question: Is the manufacturing facility at a given address currently operating, and who owns it?

Step	Category	Specific action	What it settles
1	Maps	Determine county, municipality, and special districts for the address	Which records systems to query
2	Government records	Assessor and recorder: parcel ownership, transfer history	Legal ownership of the real property
3	Corporate filings	Registry lookup on the owning entity, plus entity expansion	Who controls the owning entity
4	Government records	Building permits, code enforcement, environmental and safety permits	Whether an operating permit exists and is current
5	Public databases	Safety and environmental enforcement inspections at the site	Whether inspectors have been there, and when

Step	Category	Specific action	What it settles
6	Satellite imagery	Recent imagery, plus comparison against two years prior	Physical state: vehicles, activity, expansion, dereliction
7	Social media and internet	Job postings for that location; company site location page	Hiring signal and current claims
8	Archived websites	Prior versions of the location page	Whether the claim is new
9	Newspapers	Local coverage and legal notices	Closure, sale, dispute, or expansion
10	Court records	Litigation involving the owning entity and the property	Disputes, liens, foreclosure

Ten steps, ten categories, and note that not one of them is "search the company's name and read the first page of results." That is the difference this chapter is trying to make.

Part 2: Comparison Tables

2.1 Primary vs. Secondary vs. Tertiary Sources

Dimension	Primary	Secondary	Tertiary
Definition	Original, first-hand record created at or near the time by a participant, system, or official process	Description, interpretation, or analysis of primary material	Compilation or index of secondary material
Distance from the event	Zero to one step	Two steps	Three or more steps
Examples	Court filing, deed, corporate filing, sensor data, photograph at the scene, testimony transcript, statute text, raw imagery	News article, academic paper analyzing filings, documentary, textbook chapter on a case	Encyclopedia entry, aggregator summary, database of article abstracts
Added value	Nothing added; the record itself	Context, comparison, synthesis, expertise	Discovery and orientation
Added risk	Reflects one perspective or one instrument	Compression, misreading, framing, error propagation	Compounded error, staleness

Dimension	Primary	Secondary	Tertiary
Reliability driver	The process that created it	The rigor of the describer	The rigor of the compiler
Best use	Establishing that a specific fact is on the record	Understanding meaning, history, and comparison	Finding out what exists and where to look
Typical error	Assuming primary means true	Citing it as though it were the underlying record	Citing it at all in a professional product
Citation practice	Cite directly	Cite, and retrieve the primary it relies on	Use to find sources; do not rest a judgment on it
Weight in a product	Highest	Moderate, with the primary retrieved where possible	Orientation only

The rule: every claim you intend to rely on gets walked back to its primary source. Most open source errors are secondary-source errors.

2.2 Structured vs. Unstructured Data

Dimension	Structured	Semi-structured	Unstructured
Form	Defined fields with consistent types	Consistent containers with free-form content	Free-form
Examples	Corporate registries, dockets, permit databases, financial statement tables, vessel and aircraft feeds, parcel data	Email with headers, HTML pages, documents with consistent headings, filings with standard sections	News text, books, video, audio, photographs, social posts, scanned documents
Search method	Field query with defined operators	Parse the structure, then search the content	Keyword, then human reading
Volume tolerance	Very high; millions of rows are routine	Moderate	Low; a human must read it
Comparability	High; can be counted and aggregated	Moderate	Low until extracted
Coverage clarity	Explicit; the schema tells you what is and is not captured	Partially explicit	Opaque; you cannot know what is missing
Reproducibility	Very high; the same query returns the same result	Moderate	Low; ranked search results vary
Characteristic error	Misreading a field	Assuming structure	Misreading meaning, or

Dimension	Structured	Semi-structured	Unstructured
	definition	where it is inconsistent	missing an absence
Analytic goal	Answer definitively within a bounded scope	Extract into structure	Extract the few relevant items, then verify

The habit: convert unstructured questions into structured ones wherever a system exists to answer them.

2.3 Reliable vs. Unreliable Sources

Reliability is a property of the production process, not of familiarity, popularity, or medium.

Indicator	Reliable	Unreliable
Obligation to be accurate	Legal duty, penalty for misstatement, professional licensure	None
Review before publication	Peer review, editorial review, audit, judicial review	None
Named accountability	Identified author, filer, or custodian	Anonymous or pseudonymous with no track record
Sourcing transparency	Specific sources cited and retrievable	"Reports say," "sources tell us," no citation
Correction practice	Visible, dated corrections	No corrections ever, or silent edits
Distance from the event	Participant, system, or official record	Aggregation of other coverage
Independence of interest	No stake in the conclusion, or a declared stake	Undeclared stake; content serves the publisher's interest
Consistency with the record	Checks out against primary material	Conflicts with, or cannot be checked against, primary material
Coverage disclosure	States what it covers and does not	Implies completeness
Reproducibility	Another person can retrieve the same thing	Cannot be located again

Two cautions. First, reliability is a spectrum applied to a specific claim, not a badge applied to an outlet: a reliable outlet carrying an unsourced claim yields low credibility for that claim. Second, official sources are not automatically reliable about themselves. A government report on its own performance and a company filing describing its own prospects are both produced by interested parties, and both should be rated on the process (compelled, audited, penalized) rather than on the assumption

of good faith.

2.4 Traditional Media vs. Digital Sources

Dimension	Traditional media (print, radio, television)	Digital sources (web, social, databases, archives)
Production process	Editorial selection, fact-checking practice, correction norms	Ranges from audited compelled filings to no process at all
Speed	Hours to days	Seconds
Depth	Higher per item; space and time constraints force selection	Unlimited length, wildly variable quality
Geographic coverage	Strong local coverage historically, now collapsing with local outlet closures	Strong where connectivity and platform adoption are strong
Historical depth	Excellent; archives run back a century or more	Thin before the mid-1990s; link rot degrades even recent material
Searchability	Poor without digitization; microfilm and tape require manual work	High, but ranked and personalized rather than exhaustive
Structure	Unstructured	Both, and the structured portion is the valuable part
Persistence	Physical copies persist; broadcast is often unpreserved	Mutable and deletable; archives partially mitigate
Verification burden	Moderate; the outlet did some of the work	High; you do all of the work
Deception exposure	Lower barrier is publication cost	Publication is free, so deception is cheap
Attribution	Named bylines and identified outlets	Frequently anonymous
Cost to access	Subscription, library, purchase	Mostly free at the surface; fee for some deep web systems
Best use	Historical work, local detail, on-record statements, legal notices	Current events, structured records, technical facts, imagery, change over time
Characteristic failure	Wire reprints mistaken for independent corroboration	Circular amplification, synthetic content, ranked results mistaken for completeness

2.5 Surface Web vs. Deep Web vs. Dark Web

Dimension	Surface web	Deep web	Dark web
Definition	Crawlable and indexed by general search engines	On the ordinary internet but not indexed: behind queries, forms, logins, paywalls, or crawl exclusion	Requires overlay network client software; addresses do not resolve in ordinary DNS
Access	Any browser; findable by search	Any browser; you must go to the system and query it	Specialized client software
Relative size	The smaller portion of web content	The larger portion	Very small in comparison
Typical content	Company sites, news, blogs, public documents that are linked	Court case systems, library catalogs, corporate registries, academic databases, agency portals, subscription archives, webmail	Criminal markets, leak and extortion sites, forums, privacy-preserving mirrors of legitimate services
Authoritativeness	Mixed	Highest concentration of authoritative primary material in this chapter	Lowest; participants are adversarial and routinely defraud each other
Legality of access	Lawful	Lawful, subject to terms of service and authorization	Network use is lawful in most jurisdictions; much of the content is not, and mere viewing of some content is criminal
Discoverability	High	Requires knowing the system exists	Very low; no complete index exists
Role in this course	Starting point	Where most of your work happens	Studied conceptually only; access prohibited under Rule 4
Common confusion	Believing search results equal what exists	Confusing it with the dark web	Assuming it is only criminal, or assuming it is vast

2.6 Public Records by Source Type

Record type	Level	Typical content	Access	Latency	Primary investigative use
Deeds and property transfers	County	Grantor, grantee, date, consideration, legal description	Recorder portal or in person; often free to view, fee to copy	Days to weeks	Ownership, transfer history, encumbrances
Assessment and tax rolls	County	Assessed value, owner of record, parcel characteristics	Assessor portal, usually free	Annual	Ownership, property characteristics, delinquency
Building permits	City or county	Applicant, contractor, scope, valuation, inspection status	Permit portal, usually free	Days	Whether construction is authorized and completed
Code enforcement and violations	City or county	Complaints, inspections, citations, abatement	Portal or request	Weeks	Condition, compliance history
Business licenses and trade names	City, county, or state	Operating name, owner, location, status	Portal, small fee	Days to weeks	Operating identity beyond the legal entity
Corporate registration and annual reports	State or national	Entity identity, officers, agent, status, formation date	Registry portal, free to modest fee	Days	Entity identity and structure
UCC filings	State	Secured party, debtor, collateral description	Index search, small fee	Days	Lenders, encumbered assets
Professional licensing and discipline	State	License status, dates, disciplinary actions	Board portal, free	Weeks	Credential verification, sanctions
Court dockets and filings	Federal, state, local	Pleadings, motions, exhibits, orders, judgments	Varies from free portals to in-person only	Days	Disputes, findings, compelled documents
Bankruptcy	Federal	Assets,	Federal system,	Days	True financial

Record type	Level	Typical content	Access	Latency	Primary investigative use
filings		liabilities, creditors, insider transactions	per-page fee		condition, creditor relationships
Securities filings	National	Audited financials, risks, litigation, insiders, ownership	Free national system	Days to months	Company condition, structure, disclosures
Nonprofit information returns	National	Revenue, compensation, vendors, grants, related entities	Free public disclosure	Annual, with lag	Nonprofit finances and relationships
Campaign finance and lobbying	State and national	Contributors, amounts, expenditures, lobbying subjects	Free portals	Periodic filings	Relationships and influence activity
Environmental permits and enforcement	State and national	Permits, emissions, releases, violations, remediation	Free databases	Weeks to years	Site conditions, compliance history
Workplace safety inspections	State and national	Inspections, citations, penalties, narratives	Free databases	Weeks to months	Operating conditions, incident history
Food and facility inspections	County or state	Inspection dates, findings, scores	Free portals	Days to weeks	Whether a site is operating and how well
Government contracts and grants	All levels	Vendor, amount, scope, period, modifications	Free portals, check registers	Weeks	Vendor relationships and performance history
Vital records	State or county	Birth, marriage, death records	Restricted; indexes sometimes public	Weeks	Historical and genealogical work only; restricted for individuals
Vehicle, vessel,	State and	Registration,	Vehicle records	Weeks	Asset

Record type	Level	Typical content	Access	Latency	Primary investigative use
and aircraft registries	national	ownership, airworthiness, class	heavily restricted; vessel and aircraft largely open		ownership and movement
Legislative and regulatory records	State and national	Bills, hearings, testimony, rulemaking dockets, public comments	Free	Days	Policy direction, who is participating
Government audits and inspector general reports	All levels	Findings on programs and contractors	Free	Months to years	Independent findings on performance

2.7 Advantages, Limitations, and Investigative Value of Every Category

Category	Advantages	Limitations	Distinctive investigative value	Reliability (typical)
Internet (as channel)	Fast, broad, cheap; entry point to nearly everything	Ranked and personalized rather than exhaustive; surface indexing misses the authoritative layer	Orientation: tells you which authoritative system to query	Varies entirely by what is delivered
Books	Depth, historical coverage, curated endnote trails	Publication latency of a year or more; self-published looks identical to peer-reviewed	Pre-digital history and a finding aid to primary sources	Moderate to high for academic and established trade presses
Newspapers	Contemporaneous record, local detail, legal notices, deep archives	Circular reporting via wire reprints; catastrophic gaps where local outlets have closed	Historical local detail and statutory legal notices	Moderate; depends on the outlet and the sourcing of the specific claim
Academic papers	Method and measurement; defensible general claims; peer review	Long latency; rarely about your specific subject; single studies over-cited	Rates, mechanisms, and methods you can cite defensibly	High for peer-reviewed, low for preprints and predatory venues

Category	Advantages	Limitations	Distinctive investigative value	Reliability (typical)
Government records	Compelled, structured, primary, broad; strongest process of any category	Jurisdiction-bounded; digitization thins before the 1990s; absence is ambiguous	The definitive answer to legal and administrative facts	High
Court records	Adversarially tested; compelled discovery exhibits; indexed by party	Allegations misread as findings; state trial court access is poor; sealed matters invisible	Documents parties never intended to publish, plus authoritative findings	High for orders; nil for the truth of allegations
Radio	Functions when infrastructure fails; local and low-connectivity coverage	Ephemeral unless recorded; poor archival preservation	The only channel in some places and some conditions	Moderate; state broadcasting is a position statement, not a fact source
Television	Visual record; on-record statements; searchable via captions	Framing; clips circulate stripped of context; uneven preservation	High-quality visual evidence, especially pre-smartphone era	Moderate to high for footage as a record of what was in frame
Podcasts	Unusually candid, specific, long-form disclosure by insiders	No editorial process; conversational statements are easy to over-read	Voluntary detail no official document contains	Low to moderate; rate as a conversational statement
Satellite imagery	Physical ground truth; change over time; hard to dispute	Resolution, cloud cover, revisit gaps, stale consumer imagery; interpretation error	Converting a claim into a checkable exhibit	High as a record; the interpretation carries the risk
Social media	Fastest available; eyewitness material; self-disclosure	No process; deception, recycled and synthetic content; coverage bias; ephemeral	First visual material from inside an event	Low; verification burden sits entirely on you
Maps	Spatial relationships; jurisdictional determination; historical comparison	Currency varies; renderings are not the legal record	Tells you which records systems even exist for a location	High for official mapping; variable for community-contributed

Category	Advantages	Limitations	Distinctive investigative value	Reliability (typical)
Corporate filings	Compelled, penalized, audited for public companies; structured	Thin for private companies; the company's own characterization	What a company is willing to certify, as opposed to claim	High
Financial records	Audited and compelled where they exist; extensive public-sector transparency	Individual financial data is protected and off limits; comparison across entities is unsafe	Government spending, obligations, and vendor relationships	High for audited and compelled material
Public databases	Definitive within scope; structured; queryable; reproducible	Field definitions mislead; scope silence; variable historical depth	Bounded, checkable answers with a stated coverage boundary	High, conditional on reading the codebook
Archived websites	Converts a mutable surface into a dated sequence; recovers deleted content	Uneven capture; dynamic content fails to archive; interval-limited dating	Establishing what a party claimed at a specific time	High for what a capture contains; ambiguous for what it lacks
Dark web	Early breach and extortion indicators; threat actor insight	Legal, operational, and psychological risk; adversarial and deceptive content	Discovery that an organization's data has been exfiltrated	Very low; participants routinely lie, and listings are allegations
Metadata	Automatically generated, therefore rarely curated; establishes time, device, authorship, relationships	Editable, strippable, altered by handling; platform stripping; wrong clocks	Provenance and authenticity testing; linking files to a common origin	Moderate; corroborating rather than conclusive

Part 3: Case Studies

Each case shows how **combinations** of categories produce a finding that no single category could support. Cases marked *historical* describe real documented events, described in general terms. Cases marked *composite* are built from documented professional practice and are not real organizations; do not cite them as real.

3.1 Verifying Breaking News

A Reported Industrial Fire (composite, built from standard newsroom verification practice)

The situation. At 21:40 a video begins circulating showing a large fire at what is described as a chemical plant in a named town, with claims of a toxic release and an evacuation order. A regional newsroom must decide within an hour whether to report it, and an emergency management office needs to know whether it is real.

Categories used and the sequence.

Order	Category	Action	Result
1	Social media	Locate the earliest posting of the video, not the one you saw	Earliest identifiable post is 21:12, from an account with a two-year local history
2	Internet (reverse search)	Reverse image search on extracted frames	No earlier appearance found, which is consistent with new content but does not prove it
3	Maps	Identify the plant at the named location; determine county, municipality, and fire district	Facility exists; jurisdiction identified, which tells you which agencies to check
4	Satellite imagery / street-level	Compare fixed features in the video against street-level imagery of the site frontage	Building profile, fence line, and a distinctive tank arrangement match
5	Metadata	Examine the video file where obtainable	Metadata stripped by the platform, which is expected and therefore uninformative
6	Government records	Check the facility's environmental permits and chemical inventory reporting	Facility is permitted to store two of the substances being claimed, and not the third
7	Radio	Monitor local broadcast and, where lawful, public safety transmissions	Local station is carrying a shelter-in-place advisory for a defined area

Order	Category	Action	Result
8	Government records	Check the emergency management agency's official notification channel	Advisory confirmed, with a smaller affected area than the social claims described
9	Public databases	Check the facility's safety inspection and violation history	Two prior citations relating to storage practice
10	Newspapers	Search local coverage for prior incidents at the site	One prior fire eight years earlier

The finding. Fire confirmed at the named facility, with high confidence, based on geolocation against fixed features and confirmation from an official channel. The claimed release of one specific substance is **not supported**, because the facility is not permitted to store it, and the claimed evacuation area is larger than the official advisory. Assessment: an authentic event with accurate core facts and inaccurate embellishment, a pattern typical of fast-moving social reporting.

Methodology lessons.

Geolocation resolved authenticity; records resolved the details. Two different categories answered two different questions, and neither could have answered both. This is the core lesson of the chapter.

Stripped metadata is not a red flag. Platforms strip it routinely. An analyst who treats its absence as suspicious will be wrong most of the time.

Permits bound the claim. A chemical inventory filing is a compelled record of what is legally allowed on site. That converted a rumor into a testable proposition in one step.

The embellishment pattern is the norm. Real event, real core, inflated periphery. Reporting "confirmed, with these specific claims unsupported" is more useful than either confirming or debunking the whole.

3.2 Investigating a Company

Undisclosed Related Parties at a Municipal Vendor (composite)

The situation. A city is preparing to renew a \$6M annual waste-hauling contract. A council member asks whether the incumbent vendor is as independent of its subcontractors as it represents.

Categories used.

Category	Action	Finding
Corporate filings	Registry lookup on the vendor; enumerate officers, agent, and	Two managers, commercial registered agent

Category	Action	Finding
	formation details	
Corporate filings	Search the registry by officer name rather than by entity name	Four additional active entities share one manager
Government records	County trade name filings	Two of the four operate under trade names that appear in the vendor's subcontractor list
Government records	Property records for the vendor's stated yard address	Parcel owned by one of the four entities
Financial records	City check register and contract awards	Payments to two of the four entities under separate contracts
Court records	Docket search on all six entity names in two counties and the federal district	One collections matter, and one contract dispute naming two of the entities as co-defendants, which establishes their relationship on the record
Public databases	State environmental and transportation carrier records	Two entities share a fleet address and one shares vehicle registrations
Archived websites	Prior versions of the vendor's site	An earlier version listed the four entities as "affiliates"; the current version does not
Newspapers	Legal notices	A fictitious business name notice tying one trade name to a named principal

The finding. The vendor's subcontractors include entities under common management, contradicting the representation of independent subcontracting, assessed with high confidence on the basis of registry filings, property ownership, shared vehicle registrations, a court filing naming them as co-defendants, and the vendor's own prior website language. The finding is stated as a structural fact about common management and does not allege any legal violation, which would depend on the contract terms and applicable procurement rules.

Methodology lessons.

Search the registry by officer, not only by entity. This is the single highest-yield move in corporate research and almost nobody does it.

The archived website was the vendor's own admission. A prior version described the relationship the current version conceals. This is what makes archives a category rather than a convenience.

Six independent categories agreed. No two could be manipulated together at reasonable cost, which is what justifies high confidence, following the convergence principle from Chapter 2.

Word the finding narrowly. "Common management, contrary to the representation" is defensible. "Fraud" is not, and asserting it would be both wrong and actionable.

3.3 Locating Historical Information

Reconstructing a Century of Land Use at an Industrial Parcel (composite, using a standard environmental due diligence methodology)

The situation. A developer plans housing on a five-acre urban parcel. Prior industrial use is suspected but the chain of use before 1970 is undocumented in any current record. The question determines whether costly subsurface investigation is warranted.

Categories used, in chronological reconstruction order.

Category	Specific source	What it established
Maps	Historical fire insurance maps, which recorded building footprints, construction materials, and occupancy by name at roughly decadal intervals from the late 1800s	1898: lumber yard. 1912: a plating works occupying the northern half. 1930: expanded plating and a machine shop
Government records	City directories, which listed businesses by street address year by year	Continuous occupancy names, filling the gaps between map editions
Newspapers	Local paper archives, searched by address and by occupant name	A 1937 article describing a fire at the plating works; a 1954 legal notice of a lien; a 1961 article about a discharge complaint
Satellite imagery	Historical aerial photography from government archives, available for many areas from the 1930s onward	Structure footprints, a surface impoundment visible from 1948 to 1966, and demolition between 1971 and 1974
Government records	Building and demolition permits	Confirmed demolition dates and identified the demolition contractor
Government records	Property records: deeds and transfers	Ownership chain, including a 1968 transfer to a holding entity
Corporate filings	Registry records for that holding entity	Dissolved in 1979; officers identified
Public databases	Environmental agency site inventories and enforcement records	No listing, which is a coverage finding rather than a clean bill of health, since the operations predate the reporting regimes

Category	Specific source	What it established
Books	A local industrial history published by a county historical society	A chapter on the plating industry describing typical waste practices of the period
Court records	County index	A 1963 nuisance suit filed by neighboring owners

The finding. Continuous industrial use from approximately 1898 to 1971, including metal plating with a visible surface impoundment for at least eighteen years, assessed with high confidence. Absence from modern environmental databases is explained by the operations ending before the relevant reporting regimes existed. Recommendation: subsurface investigation targeting the mapped impoundment footprint and the northern plating area.

Methodology lessons.

Absence from a modern database means the database's scope, not the world's history. This is the coverage trap from Section 1.6.5, and it is the reason this investigation was necessary at all.

Historical maps and city directories are the backbone of pre-digital work. They are structured, dated, address-indexed, and held by libraries and archives rather than by anything a search engine indexes.

Historical aerial photography is widely available and widely unknown. Government archives hold repeated coverage of much of the developed world from the 1930s or 1940s onward, and it is the only way to see a feature that was demolished before anyone thought to record it.

Ten categories, one answer. No single source could have produced this. The chronology emerged from interleaving them.

3.4 Analyzing a Disaster

The Champlain Towers South Collapse, 2021 (historical, described at the institutional level)

The situation. In June 2021 a residential tower in Surfside, Florida partially collapsed, killing 98 people. Within days, journalists and engineers sought to establish what had been known about the building's condition beforehand, and what records existed.

Categories used by the investigating reporters and analysts.

Category	Material
Government records	Municipal building department files, including a 2018 structural engineering report that documented major structural damage to the pool deck and garage area and estimated substantial

Category	Material
	repair costs
Government records	Building department correspondence and inspection history, plus the 40-year recertification process documentation
Government records	Association and permit filings, including the eventual repair permit and its timing relative to the report
Court records	Prior litigation involving the association and contractors
Corporate filings	Entity records for the association, management, and contractors
Newspapers	Contemporaneous local coverage of the association's disputes over assessments and repairs
Satellite and aerial imagery	Comparison of imagery across years documenting adjacent construction and site conditions
Academic papers	Engineering literature on corrosion, structural fatigue, and coastal reinforced concrete deterioration, supplying the mechanisms
Metadata and documents	Document properties and version information on published reports and correspondence
Public databases	Environmental and geological data, including subsidence and groundwater studies of the area

The methodology. The critical move was ordinary and unglamorous: **the 2018 engineering report was a public municipal record, and someone requested and read it.** Once that document was public, everything else became a matter of building a timeline: what was known, when, who was told, what was authorized, and how long the repair process took relative to the documented deterioration. Records supplied the timeline; academic literature supplied the physical mechanism; imagery supplied context; litigation records supplied the history of disputes.

Lessons learned.

The decisive document was sitting in a municipal file. Nothing exotic, nothing leaked, nothing requiring special access. This is the most repeated finding in disaster investigation: the warning was in a public record that nobody had asked for.

Records establish the timeline; literature establishes the mechanism. Neither substitutes for the other. A timeline without a mechanism cannot explain; a mechanism without a timeline cannot attribute.

A standing requirement would have surfaced it earlier. Applying Chapter 1's framework: no one had a requirement reading "identify buildings in our jurisdiction with engineering reports documenting

structural deficiency and no completed repair permit." That query is answerable from records the municipality already held.

Handle the human dimension carefully. An institutional analysis of records, decisions, and engineering is appropriate. Compiling information about victims or residents as individuals is not, and the distinction should be explicit in any product.

3.5 Researching Legal Proceedings

Litigation Document Archives as an Open Source (historical pattern)

The situation. Researchers and journalists studying corporate conduct in the tobacco, pharmaceutical, chemical, and firearms industries have relied heavily on documents produced in discovery and entered into public court records, and in several instances made available in large searchable public archives maintained by universities and public interest organizations.

Why this pattern matters. Discovery compels the production of internal material: memoranda, research results, marketing plans, correspondence, and studies never intended for publication. When those documents are attached as exhibits or admitted at trial, they enter the public record. Large-scale digitization has turned some of these collections into searchable archives containing millions of pages of primary corporate documents.

The methodology for using litigation records analytically.

Step	Action	Category
1	Identify the litigation: which cases, which courts, which years	Court records
2	Retrieve the docket and read it as an index; identify filings likely to carry exhibits, such as summary judgment motions and trial exhibit lists	Court records
3	Retrieve the exhibits, which is where internal documents live	Court records
4	Distinguish document types rigorously: complaint allegations, expert reports, deposition testimony, admitted exhibits, and judicial findings	Court records
5	Search dedicated document archives where they exist for the industry	Public databases
6	Corroborate document authenticity: does the docket	Court records, metadata

Step	Action	Category
	confirm the exhibit, and does the document's metadata and internal detail fit its claimed origin	
7	Place documents in context using contemporaneous coverage and the academic literature that has already analyzed them	Newspapers, academic papers
8	Check the appellate history, since findings can be reversed	Court records

Lessons learned.

Exhibits are the prize; the docket is the map. New researchers read complaints because they are easy to find and quotable. Practitioners read the docket to locate exhibits.

Document type determines evidentiary weight, absolutely. An admitted exhibit is different from an allegation quoted in a complaint, which is different from an expert's contested opinion, which is different from a judicial finding. Conflating them is the most common error in this category and the one with the most legal exposure.

Settlements resolve disputes without establishing facts. A settled case tells you the parties agreed to end litigation. It does not tell you what happened.

Access is unequal. Federal dockets are reasonably accessible; state trial dockets often are not. Any product must state which systems were searched, because a clean federal search says very little.

3.6 Identifying Misinformation

Recycled Imagery and Coordinated Amplification (composite, built from documented patterns)

The situation. During a flood, images and video circulate purporting to show conditions in an affected town: a submerged downtown, a wild animal in floodwaters, and a collapsed bridge. Local officials are being asked to respond, and each item must be assessed separately.

Category-by-category analysis of the three items.

Item 1: the submerged downtown.

Category	Action	Result
Internet (reverse search)	Reverse image search	Same image appears in coverage of a different flood four years earlier, in another country
Maps	Compare visible signage and	No match; street grid and

Category	Action	Result
	street layout against the claimed town	language on signage inconsistent
Metadata	Examine file where available	Stripped

Assessment: recycled authentic imagery, misattributed. High confidence, on the basis of a documented earlier appearance.

Item 2: the animal in floodwaters.

Category	Action	Result
Internet (reverse search)	Reverse search on the composite and on the cropped animal region separately	The animal region matches a stock wildlife photograph; the street background matches a genuine photograph from the current flood
Metadata	Examine file	Editing software identified in the producer field, with a modification timestamp after the flood began

Assessment: manipulated composite. High confidence. Note that searching the cropped region separately was the step that resolved it, because a composite may not match anything as a whole.

Item 3: the collapsed bridge.

Category	Action	Result
Internet (reverse search)	Reverse search	No earlier appearance found
Maps and satellite imagery	Geolocate against the claimed location	Bridge structure and approach match the claimed site
Government records	Check the state transportation agency's public bridge inventory and any closure notices	Bridge listed; a closure notice was issued the previous day
Radio	Local broadcast	Closure confirmed on air

Assessment: authentic and accurately attributed. High confidence.

The amplification analysis.

Category	Observation
Social media	Items 1 and 2 were posted by accounts created within the past month, posting near-identical text

Category	Observation
Social media	Posting activity clustered in hours inconsistent with the claimed local time zone
Newspapers	Two outlets carried item 1 without independent verification, citing "social media reports"
Archived websites	One site that published item 1 had been registered eleven days earlier

Lessons learned.

Assess items individually. Two of three were false and one was true. An analyst who dismissed the whole set, or accepted the whole set, would have been wrong either way. This is the discipline that most distinguishes analysis from advocacy.

Search cropped regions separately. Composites defeat whole-image search.

Official records resolved the true item fastest. A transportation agency's bridge inventory and closure notice settled in two minutes what social media analysis could not settle at all.

Registration dates of the publishing sites were as informative as the content. Infrastructure records, from Section 1.16, are a misinformation-analysis tool.

3.7 Supporting a Cybersecurity Investigation

A Vendor Breach Reaching a Client (composite)

The situation. A hospital system's security team learns that one of its billing vendors has been named on a ransomware group's leak site. The vendor has said only that it is "investigating an IT incident." The security team must determine whether patient data is exposed and what to do this week.

Categories used.

Category	Action	Finding
Dark web (via monitoring service, not direct access)	Review the leak site listing details supplied by the monitoring vendor	Listing names the vendor, claims a data volume, and includes a small sample file index
Internet, corporate filings	Confirm the entity named on the listing is in fact the hospital's vendor and not a similarly named company	Confirmed via registry records and the vendor's own site
Archived websites	Retrieve prior versions of the vendor's client list and service description pages	An earlier version listed the hospital by name and described the specific data types processed, which the current version no

Category	Action	Finding
		longer does
Government records	Check the state and federal breach notification portals	No notification filed as of the review date, which is a finding about the vendor's compliance posture
Public databases	Certificate transparency, passive DNS, and registration data for the vendor's domains and any lookalikes	Two lookalike domains registered during the incident window
Public databases	Internet-wide scan data for the vendor's address space	An exposed remote access service, present in scan history for several months
Metadata	Examine the sample document index and any published sample documents provided through the monitoring service	File paths in the index reveal a directory structure naming client folders, one of which matches the hospital's abbreviation
Corporate filings	Review the vendor's ownership and any recent changes	Recent change of control, which raises questions about integration and control continuity
Court records	Docket search on the vendor	Two prior contract disputes, one alleging inadequate security practices
Government records	Retrieve the hospital's own contract with the vendor from public procurement records, since the hospital is a public entity	Contract specifies notification timelines and audit rights the hospital can now invoke

The finding. Exposure of hospital data is **likely** (roughly 70%), assessed with moderate confidence: the leak site index contains a directory matching the hospital's identifier, and archived vendor pages confirm the hospital's data was processed by the vendor and describe the data types. Confidence is moderate rather than high because leak site claims are made by adversaries who lie, false listings occur, and the index has not been independently corroborated. Recommended immediate actions include invoking the contractual notification and audit provisions, preparing regulatory notification on the assumption of exposure, rotating any shared credentials, and monitoring the two lookalike domains.

Lessons learned.

The archived vendor website was the pivot. The current site had removed the client list and the data-type description. The archive established both that the hospital was a client and what data was involved, which the vendor was not saying.

File paths are metadata. A directory listing is a structural record of how an organization stores data,

and client folder names are frequently visible in it.

A leak site listing is an allegation. Ransomware operators inflate, misattribute, and occasionally fabricate. The correct posture is to plan for exposure while stating that the claim is uncorroborated.

Public procurement records gave the hospital its leverage. The contract itself was a public record, and it contained the notification and audit rights that turned an intelligence finding into an action.

No direct dark web access was required. Everything came through a monitoring service, archives, registries, and public records. This is how the category is handled professionally.

3.8 Cross-Case Comparison

Case	Categories that carried the finding	Decisive move	Lesson
Industrial fire verification	Social media, street-level imagery, environmental permits, radio	Geolocation for authenticity, permits for the details	Different categories answer different questions
Municipal vendor	Corporate registry, property records, court records, archived website	Searching the registry by officer name	Structured queries beat keyword searching
Historical parcel	Historical maps, city directories, newspaper archives, historical aerals	Interleaving decadal map editions with annual directories	Absence from a modern database is a scope statement
Champlain Towers South	Municipal building records, court records, academic literature	Requesting and reading a public engineering report	The warning is usually in a record nobody asked for
Litigation archives	Court dockets and exhibits, document archives	Reading the docket to locate exhibits	Document type determines evidentiary weight
Flood misinformation	Reverse search, maps, transportation records, registration data	Assessing each item separately; searching cropped regions	Mixed sets are the norm; judge item by item
Vendor breach	Monitoring service, archived website, scan and certificate data, procurement records	Archived client list and data-type description	Archives recover what an organization has quietly removed

Two patterns recur in all seven. **The decisive source is rarely the obvious one**, and it is frequently a structured official record that a general search engine does not index. And **every finding rests on**

agreement across categories with unrelated failure modes, which is what makes it survive scrutiny.

Part 4: Chapter Summary

A source category is not a search tool. Categories are defined by who produces the information, through what process, and under what obligation. The internet, web archives, and the dark web are access layers, not content types. Plan collection by category, then choose tools. Reversing that order is the most common structural mistake in open source work.

Six properties describe every category: content, production process, access model, coverage, latency, and failure mode. Production process is the one that determines reliability. A corporate filing is reliable because it is compelled, formatted, reviewed, and penalized, not because companies are honest. A social media post is unreliable because there is no process at all.

Primary, secondary, and tertiary determines what a finding is worth. Every step from the event adds error, compression, and interpretation. Walk every claim you rely on back to the primary source. But primary means closer to the event, not true: a police report, a company filing, and an official statement are all primary and all written by interested parties.

Structured beats unstructured when a structured system exists. Field queries produce bounded, reproducible, checkable answers with explicit coverage. Keyword searching produces ranked, personalized, unbounded results. Convert unstructured questions into structured ones whenever a system can answer them, and always read the codebook before using a field.

The published and edited categories (books, newspapers, academic papers, radio, television, podcasts) supply depth, history, local detail, method, on-record statements, and candid disclosure. Their common weaknesses are latency, framing, uneven preservation, and circular reporting through wire reprints and aggregation. Newspaper archives and legal notices, books' endnote trails, theses on narrow subjects, and closed-caption archives are the specific high-value assets in this group.

The official and compelled categories (government records, court records, corporate filings, financial records, public databases) are the backbone of serious open source work. They are primary, structured, and produced under legal obligation. Their characteristic trap is coverage: a missing record is a statement about the record system's jurisdiction, scope, digitization, and naming, not about the world. In court records specifically, a complaint is an allegation, an order is a finding, and a settlement establishes nothing.

The observational and technical categories (satellite imagery, maps, metadata) supply physical and system-generated ground truth that no one curated. Imagery converts claims into checkable exhibits, and change detection across dated captures is its most valuable technique. Maps determine jurisdiction, which determines which records exist. Metadata establishes time, device, authorship, and relationships, and it is corroborating rather than conclusive, because every field can be edited, stripped, or rewritten by ordinary handling.

Social media is the fastest category and the one with no production process. Its value is eyewitness

material, geolocatable visual content, and voluntary disclosure. Its verification burden is total, and it is where the ethical rules bind most tightly, because compiling a profile of a private individual from public posts is trivially easy and is the default outcome of unstructured browsing.

Archived websites are a category, not a convenience. They convert a mutable surface into a dated, citable sequence, establish what a party claimed at a specific time, recover deleted content, and expose quiet revision. In three of this chapter's seven case studies, the archived version of a website was the pivot.

The surface web, deep web, and dark web are three different things. The deep web is where nearly all authoritative material lives and it is entirely mundane. The dark web is a small overlay-network layer holding criminal markets, leak and extortion sites, and legitimate privacy-preserving services. It matters analytically for breach and credential discovery, it is handled professionally through monitoring services rather than browsing, and you will not access it in this course.

Corroboration across categories is worth far more than corroboration within one. Three articles agreeing may be one wire story. A permit, an image, and a filing agreeing is strong, because the three have unrelated production processes and unrelated failure modes and cannot be coordinated cheaply. State, for each significant finding, which categories support it.

Say what you searched and what you did not. Every negative finding carries its coverage boundary. "No records returned from these two systems for these six name variants on this date" is a professional finding. "I could not find anything" is not a finding at all.

And the ethics carry forward unchanged. No private individuals. Public means genuinely public. Document everything. No dark web access. The categories in this chapter are powerful precisely because they are compelled, permanent, and cross-referenceable, and that power is exactly why the constraints matter.

Part 5: Key Terms

Grouped by theme. Terms carried forward from earlier chapters are marked (Ch.1) or (Ch.2) and restated briefly.

5.1 Source Framework

Source category. A class of information defined by who produces it, through what process, and under what obligation. Distinct from a tool, which is a means of searching one or more categories.

Production process. The set of obligations, formats, reviews, and penalties under which a source is created. The real determinant of reliability.

Access model. How a source is obtained: free, fee, registration, query, formal request, or in person.

Coverage. The jurisdictions, dates, languages, populations, and subjects a source includes, and by implication those it omits.

Latency. The interval between an event and the availability of a record of it.

Coverage boundary. The explicit statement of what was and was not searched, attached to every negative finding.

Primary source. An original first-hand record created at or near the time by a participant, a system, or an official process.

Secondary source. A description, interpretation, or analysis of primary material.

Tertiary source. A compilation or index of secondary material.

Structured data. Information organized into defined fields with consistent types.

Unstructured data. Free-form information: text, audio, video, images.

Semi-structured data. Consistent containers with free-form content, such as email or HTML.

Codebook / data dictionary. The documentation defining what each field in a database means. Reading it prevents the most common database error.

5.2 Web and Access Layers

Surface web. Pages that general search engines can crawl and index.

Deep web. Content on the ordinary internet that is not indexed, because it sits behind a query, form, login, or paywall, or is excluded from crawling. Where most authoritative material lives.

Dark web. Content requiring specific overlay network client software to reach, using addresses that do not resolve in ordinary DNS.

Overlay network. A network built on top of the internet that routes traffic through intermediaries to obscure origin and destination.

Leak site / extortion site. A site operated by a ransomware or extortion group publishing victim names and stolen data to pressure payment.

Ranked and personalized results. The property of search engines that different users, and the same user at different times, receive different result sets. Requires that queries, engines, dates, and results be documented.

Search operator. A syntax element restricting a query by site, file type, date range, or exact phrase.

5.3 Records Categories

Government record. A record created or held by a public body, frequently public by statute.

Public records request. A statutory mechanism (freedom of information or public records acts) by which any member of the public may compel disclosure of records, subject to exemptions.

Exemption. A statutory basis for withholding or redacting a record, such as privacy, ongoing investigation, deliberative process, trade secret, or security.

Docket. The numbered index of every filing in a court case, with dates. The starting point for any court records work.

Complaint / petition. The document initiating a civil case, containing one party's allegations. Not evidence and not findings.

Answer. The responding party's denials and defenses.

Exhibit. A document attached to a filing, frequently internal material produced in discovery. Often the most valuable content in a case file.

Deposition. Sworn out-of-court testimony, taken under oath.

Order / opinion. A judicial decision, containing the court's actual findings and reasoning.

Judgment / disposition. The outcome of a case: verdict, settlement, dismissal, or default.

Settlement. A negotiated resolution that ends a dispute without establishing facts.

Sealed record. A record withheld from public access by court order; invisible to open source research by design.

Discovery. The pre-trial process compelling parties to produce documents and information.

Deed. A recorded instrument transferring an interest in real property.

Assessment roll. A county record of assessed property values and owners of record.

Building permit. An authorization to construct or alter, with applicant, contractor, scope, valuation, and inspection status.

Code enforcement record. A record of complaints, inspections, and citations regarding property condition or use.

Trade name / DBA filing. A registration of an operating name different from a legal entity name, frequently filed at county level.

Foreign qualification. Registration permitting an entity formed in one jurisdiction to do business in another.

Annual report (registry). A periodic filing updating an entity's officers, address, and status.

Periodic report (securities). A public company's annual or quarterly filing containing audited or reviewed financials, risk factors, legal proceedings, and management discussion.

Event report (securities). A filing disclosing a material event such as a leadership change, acquisition, default, or auditor change.

Risk factors. The section of a public filing disclosing what could go wrong. Changes between years are the analytic signal.

Related-party transaction. Business conducted with insiders or their affiliates, disclosed in filings.

Management discussion and analysis. Management's narrative explanation of results, frequently

containing customer and supplier concentration disclosures.

UCC filing. A public notice of a security interest in personal property, identifying secured party, debtor, and collateral.

Lien. A recorded claim against property securing a debt or obligation.

Bankruptcy schedules. Filings listing assets, liabilities, creditors, and insider transactions.

Nonprofit information return. A public annual tax filing by a nonprofit disclosing revenue, compensation, vendors, grants, and related organizations.

Beneficial ownership. The natural persons who ultimately own or control an entity.

Entity expansion (Ch.2). Enumerating all name variants, former names, subsidiaries, agents, officers, and jurisdictions before beginning record searches.

Continuing disclosure. Ongoing financial reporting obligations attached to municipal bond issuance.

Check register. A published list of individual payments made by a government body to vendors.

Debarment list. A public list of parties excluded from receiving government contracts.

5.4 Media and Publication

Legal notice. A statutorily required publication in a newspaper of record, covering foreclosures, probate, name changes, licenses, procurement, and similar matters.

Newspaper of record. A publication designated for statutory legal notice publication in a jurisdiction.

Wire service. An organization supplying identical stories to many outlets. Multiple appearances of one wire story are one source.

Byline and dateline. The reporter's name and the location and date of reporting. A datelined byline indicates presence.

Correction record. An outlet's practice of publishing dated corrections; an indicator of process.

Peer review. Evaluation of a manuscript by independent specialists before publication.

Preprint. A manuscript posted publicly before peer review.

Predatory venue. A journal or conference that publishes for a fee without meaningful review.

Systematic review / meta-analysis. A study synthesizing many prior studies; worth more than any single study.

Retraction. Formal withdrawal of a published paper. Retracted work continues circulating and should be checked.

Gray literature (Ch.2). Publicly available material that is not formally published or indexed.

City directory. A historical annual publication listing residents and businesses by name and address. A backbone source for pre-digital research.

Fire insurance map. A historical map series recording building footprints, materials, and occupancy by name, at roughly decadal intervals.

Closed-caption archive. A searchable text record of broadcast content, used to locate footage by keyword.

5.5 Technical and Observational

Satellite imagery types. Optical (visible light), multispectral (additional bands), thermal or infrared (heat), and synthetic aperture radar (sees through cloud and at night).

Revisit rate. How frequently a satellite images the same location.

Change detection (Ch.2). Comparison of imagery of the same location across dates to identify what changed and when.

Capture date. The date an image was taken, as distinct from the date it was published or accessed. Must be established before any imagery-based claim.

Ground sample distance / resolution. The size of the smallest distinguishable feature; determines what may be claimed.

Chronolocation (Ch.2). Determining when content was captured, from shadows, weather, or dated features.

Geolocation (Ch.2). Determining where content was captured, by matching fixed features.

Cadastral / parcel data. Mapped property boundaries with ownership references.

Zoning. The regulatory designation of permitted land uses in an area.

Special district. A local governmental unit (water, fire, school, utility) with its own records.

Metadata. Data about data; information a file or system records about content, separately from the content.

EXIF. The metadata standard for photographs, carrying camera, settings, timestamps, and sometimes GPS.

IPTC / XMP. Additional metadata standards used for descriptive and rights information in images and documents.

Producer / creating application. Metadata fields identifying the software that generated a file.

Total editing time. An office document field recording accumulated editing duration.

Revision number. An office document field recording how many times a file was saved.

Tracked changes. Recorded edits within a document, frequently left in published files.

Email header. The relay and authentication record of a message, including originating server, path, timestamps, and message identifier.

Filesystem timestamps. Created, modified, and accessed dates recorded by the operating system.

Metadata stripping. Removal of metadata, performed deliberately before publication or automatically by platforms on upload.

File path disclosure. Directory structures visible in documents, archives, or index listings, frequently revealing internal organization and usernames.

5.6 Analytic and Ethical

Source reliability (Ch.1). Assessment of a provider's track record, rated A through F.

Information credibility (Ch.1). Assessment of a specific claim's plausibility, rated 1 through 6.

Corroboration (Ch.1). Independent confirmation by a source that did not derive from the first.

Cross-category corroboration. Confirmation from categories with unrelated production processes and failure modes. Worth substantially more than corroboration within a single category.

Circular reporting (Ch.1). Apparent corroboration created when multiple sources trace to one unverified origin.

Coverage gap. A portion of reality a source systematically does not record.

False negative. Concluding something does not exist because a source with limited coverage did not report it. The most dangerous error in source-based work.

Convergence of weak indicators (Ch.2). Multiple independent low-strength indicators supporting a strong conclusion, provided they could not be cheaply coordinated.

Mosaic effect (Ch.2). Creation of a sensitive whole from individually non-sensitive public parts.

Recycled content (Ch.2). Authentic material from an earlier event republished as current.

Coordinated inauthentic behavior (Ch.2). Manufactured or centrally directed content presented as independent and organic.

Allegation versus finding. The distinction between what a party claims in a filing and what a court has determined. Conflating them carries defamation exposure.

Name match versus identity match. The principle that a matching name is not proof of the same person or entity without a second corroborating identifier.

Part 6: Knowledge Assessment

6.1 Short Answer (20 questions)

Answer in three to five sentences unless otherwise noted.

1. Explain the difference between a source category and a search tool, and give one consequence of confusing them.
2. Name the six properties used to describe every category in this chapter, and explain which one

determines reliability and why.

3. Distinguish primary, secondary, and tertiary sources with an example of each, and state the working rule about relying on secondary material.
4. Explain why "primary" does not mean "true," using two examples.
5. Distinguish structured from unstructured data and explain why converting an unstructured question into a structured one improves an investigation.
6. Explain what the codebook rule is and give an example of a field whose plain-language name could mislead.
7. Distinguish the surface web, deep web, and dark web, and identify the most common confusion between two of them.
8. Explain why a search engine is a starting point and never a finishing point.
9. Describe three things books provide that no other category provides as well.
10. Explain why the legal notices section of a newspaper is an unusually valuable open source.
11. Describe four checks you would apply to a news article before relying on it.
12. Explain how to assess an academic paper without domain expertise. Name at least four checks.
13. Explain why the absence of a record in a government database is not evidence that an event did not occur. Give a concrete example.
14. Describe the structure of a court case file and explain the difference between a complaint, an order, and a settlement.
15. Explain why exhibits are often the most valuable material in court records.
16. Describe two ways radio remains analytically relevant, and state its main failure mode.
17. Explain why podcasts are a useful open source, and state how a statement made in a podcast should be rated relative to the same statement in a filing.
18. Describe the five-step discipline of change detection with satellite imagery.
19. Explain what metadata is, list four things it can establish, and state four reasons it is corroborating rather than conclusive.
20. Explain why corroboration across categories is worth more than corroboration within a category, with an example of each.

6.2 Multiple Choice (15 questions)

1. The internet is best described as: a) The largest single open source category b) A delivery channel through which many categories are reached c) A synonym for the surface web d) A tertiary source
2. Which property most determines whether a source is reliable? a) How widely known the publisher is b) Whether it appears in search results c) The production process under which it was created d)

Whether it is free to access

3. A news article describing a court filing is: a) A primary source b) A secondary source c) A tertiary source d) Structured data
4. Which of the following is a primary source? a) An encyclopedia entry about a company b) A textbook chapter analyzing a merger c) A recorded deed transferring property d) A news summary of quarterly results
5. The deep web consists primarily of: a) Criminal marketplaces b) Content requiring overlay network software c) Content on the ordinary internet that is not indexed, such as databases and case management systems d) Deleted web pages
6. In a court file, a complaint contains: a) The court's findings of fact b) One party's allegations c) The final judgment d) Sworn deposition testimony
7. Which record type would most directly establish who legally owns a specific building? a) A news article about the building b) The county recorder's deed records c) The company's marketing website d) A social media post by a tenant
8. A wire service story appearing in twenty newspapers should be counted as: a) Twenty independent corroborating sources b) One source c) Twenty primary sources d) A structured dataset
9. The most common error when using a public database is: a) Failing to pay the access fee b) Misreading a field because its definition was not checked c) Using too many search operators d) Downloading too much data
10. Historical fire insurance maps are especially valuable because they recorded: a) Satellite imagery before satellites existed b) Building footprints, construction materials, and occupancy by name at regular intervals c) Property ownership disputes d) Census demographics
11. Archived websites are best described as: a) A convenience for finding broken links b) A distinct category that converts a mutable surface into a dated, citable sequence c) A subset of the dark web d) A tertiary source
12. Which statement about metadata is correct? a) A GPS coordinate in an image proves where it was taken b) Absence of metadata in a social media image indicates manipulation c) Metadata is corroborating rather than conclusive, because fields can be edited, stripped, or rewritten by handling d) Metadata cannot be altered without specialized forensic tools
13. Professional practice for dark web material in a corporate setting is primarily: a) Direct browsing by any interested analyst b) Monitoring services, sector sharing, and law enforcement liaison, with direct collection only under written authorization on isolated infrastructure c) Ignoring the category entirely d) Purchasing data from marketplaces to assess exposure
14. Which pairing represents the strongest corroboration? a) Three articles from three outlets that all cite the same official statement b) A building permit, a satellite image, and a local news report c) Five social media posts with identical wording d) Two blog posts summarizing the same press release
15. A registry search returns no results for a company name. The correct conclusion is: a) The company

does not exist b) The company is fraudulent c) The company is not registered in that jurisdiction under that exact name, and other jurisdictions, former names, and trade names remain unchecked d) The registry is unreliable

6.3 True or False (10 questions)

Mark true or false. For each false statement, write one sentence explaining what makes it false.

1. The deep web and the dark web are two names for the same thing.
 2. A court complaint establishes that the alleged conduct occurred.
 3. Reading a database's codebook before using its fields prevents a common class of error.
 4. Absence of a record in a government database is strong evidence that the underlying event did not happen.
 5. Most litigation in the United States occurs in federal courts, so a federal docket search is generally sufficient.
 6. Consumer satellite imagery platforms always display imagery captured within the past month.
 7. Most social media platforms strip metadata from images on upload.
 8. Historical aerial photography of much of the developed world exists from roughly the 1930s or 1940s onward and is publicly available.
 9. Corroboration from three sources within the same category is analytically equivalent to corroboration from three different categories.
 10. A settlement in a civil case constitutes a judicial finding that the defendant's conduct was unlawful.
-

Part 7: Critical Thinking Exercises

For each exercise, answer the questions posed. Where you are asked to select sources, name the categories, name the specific systems or source types within them, and state what each would settle.

Selecting appropriate sources

7.1 A city council must decide whether to grant a permit to a company claiming it has operated similar facilities in four other states for at least five years. Name the four categories you would use first, the specific source within each, and what each would settle. Then name the one claim in that sentence that open sources cannot fully verify, and say why.

7.2 A nonprofit board suspects that a grantee organization has overstated the scale of its program. Which categories apply? Identify the single highest-value document and explain why.

7.3 An insurer must determine whether a warehouse claimed to have been operating at the time of a fire was in fact in use. Build a five-category collection plan, in order, and state what would make you

conclude "unresolved" rather than yes or no.

7.4 A researcher wants to know how many facilities of a specific industrial type existed in a metropolitan area in 1955. Which categories can answer this, and which cannot? Explain why the modern-database answer will be wrong.

Evaluating source credibility

7.5 Two documents make opposite claims about a company's revenue: an audited annual filing and a widely shared industry blog post citing "internal figures." Rate each for reliability and credibility, and write the sentence you would use in a product.

7.6 A well-regarded national newspaper publishes an article whose only sourcing is "people familiar with the matter." Assess reliability and credibility separately and explain how a reliable outlet can produce a low-credibility claim.

7.7 A preprint reports a striking finding that has been covered by dozens of outlets. Describe the checks you would perform before citing it, and state how you would word a reference to it in a product.

7.8 A government agency publishes a report evaluating its own program and concludes the program is effective. Is this reliable? Answer using the production-process framework rather than an assumption about good faith, and name the source that would corroborate or contradict it.

Recognizing bias and misinformation

7.9 A video circulates showing flooding in a named city. It has no earlier appearance in reverse image search, the file metadata is absent, and it is being shared by three accounts created in the past month using identical wording. Assess each of those three observations separately: what does each support, and what does each not support?

7.10 A company's website today makes no mention of a product line that industry participants recall it selling. Describe the category and method that would resolve whether the product line existed, and state the dating precision you could achieve.

7.11 A historical newspaper article from 1948 describes a factory's labor dispute in strongly partisan terms. How do you use a biased primary source? Explain what it is reliable evidence of, and what it is not.

7.12 An analyst notices that every article supporting one interpretation of an event traces back to a single think tank report, while articles supporting the other interpretation trace to two separate government datasets. Explain what this pattern means for the weight of each side, and name the failure mode at risk if the analyst ignores it.

Corroborating evidence

7.13 You have a satellite image appearing to show a new structure at a facility. Name three sources from three different categories that could corroborate the interpretation, and explain which single one would most raise your confidence and why.

7.14 You have a claim that an entity is controlled by a specific individual. Name four corroborating sources across at least three categories, and explain what would constitute a second identifier sufficient to move from "a party of the same name" to an identity assertion.

7.15 A document surfaces purporting to be an internal report. You have its metadata, its content, and nothing else. Describe how you would attempt corroboration, what you would conclude if all corroboration attempts failed, and the exact wording you would use in a product to describe an uncorroborated document.

Explaining reliability differences

7.16 Explain to a non-specialist why a county building permit is more reliable than a company's press release about the same construction project, without using the word "trust."

7.17 Explain why a podcast interview in which an executive states a customer concentration figure is useful but should be rated lower than the same figure in a periodic filing. Then explain the circumstance in which the podcast statement is the more valuable of the two.

7.18 A colleague argues that because anyone can edit a community-contributed map, such maps should never be used in an investigation. Respond, using the production-process framework, and state the conditions under which you would and would not rely on one.

Part 8: Hands-On Lab Exercises

General Lab Guidance

Rules 1 through 4 apply to every lab. No private individuals, public means genuinely public, document as you go, and no dark web access.

Rubric scale. Each criterion scores 0 to 4:

Score	Meaning
4	Professional quality; deliverable to a real consumer unchanged
3	Competent; meets requirements with minor weaknesses
2	Developing; partial compliance, notable gaps
1	Attempted; concept recognized but misapplied
0	Missing or fundamentally incorrect

Universal requirements, graded in every lab. Every judgment carries a confidence level. Every product names at least one intelligence gap. Every finding appears in a collection log with source, URL or citation, access date and time, and reliability and credibility ratings. Every negative finding carries

its coverage boundary.

Lab 1: Classifying Open Sources

Time required: 60 to 90 minutes **Format:** Individual, then group discussion **Difficulty:** Foundational

Objectives

1. Classify sources by category, by primary/secondary/tertiary type, and by structure.
2. Apply the six-property framework to unfamiliar sources.
3. Recognize that some items are access layers rather than categories.
4. Identify the characteristic failure mode of a given source.

Prerequisites

- Sections 1.1 through 1.20 and comparison tables 2.1, 2.2, and 2.5.
- No internet access required for Parts 1 and 2.

Part 1: The Classification Set (30 min)

Classify each of the thirty items below on Worksheet 1A. For each: category, primary/secondary/tertiary, structured/semi-structured/unstructured, and its single most likely failure mode.

1. A county recorder's index of deeds
2. A Wikipedia article about a company
3. A satellite image with a stated capture date of 14 June
4. A 1931 city directory listing businesses by address
5. A podcast interview with a company's chief technology officer
6. A search engine results page
7. A peer-reviewed paper on concrete corrosion rates
8. A ransomware group's leak site listing
9. A federal court docket sheet
10. A press release on a company's website
11. An archived capture of that same website from three years ago
12. A newspaper obituary from 1962
13. The EXIF data in a photograph

- 14.A state corporate registry annual report filing
- 15.A television news segment showing a fire
- 16.A textbook chapter summarizing a legal doctrine
- 17.A public safety radio transmission
- 18.A county parcel map viewer
- 19.A preprint posted to an academic server
- 20.A UCC filing index entry
- 21.A nonprofit's annual information return
- 22.A tweet from an eyewitness at an event
- 23.A municipal check register listing vendor payments
- 24.A book published by a university press
- 25.A wire service story reprinted in a local paper
- 26.A closed-caption transcript of a broadcast
- 27.A certificate transparency log entry
- 28.A zoning ordinance map
- 29.An encyclopedia of industrial history
- 30.A complaint filed in a civil case

Three of the thirty are access layers or tools rather than content categories. Identify them and explain why the distinction matters.

Part 2: Six-Property Analysis (25 min)

Select five items from the list, including at least one you found difficult to classify. Complete Worksheet 1B: fill in all six properties for each.

Part 3: Source Selection Drill (25 min)

For each of the six questions below, name the two categories you would use first, the specific source within each, and what each would settle. Then name one category that would be a waste of time for that question and explain why.

1. Who owns the building at a given street address?
2. Did a specific bridge close on a specific date?
3. What did a company claim about its client list in 2019?
4. What is the typical failure rate of a specific type of industrial valve?

5. Is a warehouse at a given address currently in use?
6. Did an executive publicly discuss the company's largest customer?

Worksheet 1A: Classification

#	Item	Category	Primary / Secondary / Tertiary	Structure	Most likely failure mode
1					
...					
30					

Access layers or tools rather than categories: _____, _____, _____. Why it matters:

Worksheet 1B: Six-Property Analysis

Item	Content	Production process	Access model	Coverage	Latency	Failure mode
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Worksheet 1C: Source Selection Drill

Question	Category 1 + specific source	Settles	Category 2 + specific source	Settles	Category to avoid + why
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Expected Outcomes

- Items 2, 16, and 29 classified as tertiary. Items 1, 3, 4, 9, 12, 13, 14, 18, 20, 21, 22, 23, 26, 27, 28, and 30 classified as primary. Items 5, 7, 8, 10, 15, 19, 24, and 25 will generate legitimate debate; the reasoning matters more than the label. A press release is primary as evidence of what a company said and secondary as evidence of what happened, and a strong answer says so.
- Items 6, 11, and 8 are the intended access-layer answers, with item 6 (a search results page) being the clearest, item 11 (an archive capture) being a delivery mechanism for content of whatever type the original page was, and item 8 sitting on the dark web access layer. Well-argued alternatives are acceptable if the reasoning distinguishes channel from content.
- Structured: 1, 9, 13, 14, 18, 20, 23, 27, 28. Unstructured: 5, 12, 15, 17, 22, 24, 29. The rest are debatable and should be argued.
- In Part 3, question 4 should route to academic papers and technical standards, and question 3 to archived websites. Students who route question 3 to social media or general search have missed the chapter's central point.
- The "category to avoid" answers should reason about production process and coverage, not

convenience.

Discussion Questions

1. Which item was hardest to classify, and what does the difficulty reveal about the framework's limits?
2. A press release can be primary or secondary depending on the question asked. Give two questions where its type differs.
3. Why does treating a search results page as a source lead to systematic error?
4. Pick one item you classified as unstructured. What structured system, if any, could answer the same question?

Grading Rubric: Lab 1 (44 points)

Criterion	Weight	4 points looks like
Category classification accuracy	x3	26 or more of 30 correct
Primary / secondary / tertiary accuracy	x2	Correct, with the debatable items argued rather than guessed
Structure classification	x2	Correct, with semi-structured recognized where appropriate
Failure mode identification	x2	Specific to the source, not generic
Access layer identification	x1	Three identified with a clear channel-versus-content explanation
Six-property analysis	x2	All six filled accurately for five items, including a difficult one
Source selection drill	x3	Specific systems named, not category labels; settles column precise
Category-to-avoid reasoning	x1	Reasons from production process and coverage

Total possible: 64 weighted. Passing: 44.

Lab 2: Multi-Category Research on a Public Topic

Time required: 2.5 to 3 hours **Format:** Individual **Difficulty:** Intermediate

Objectives

1. Research one topic across at least five source categories.

2. Compare what each category contributes, and where they agree and conflict.
3. Demonstrate cross-category corroboration and detect within-category circularity.
4. Produce a comparison analysis and a short intelligence product.

Prerequisites

- All of Part 1, and comparison tables 2.1 and 2.7.
- Internet access.
- Completion of Lab 1.

Topic Selection

Choose a **public topic about an organization, place, facility, or policy**. Not a person. Suitable topics:

- A specific public infrastructure project in your region (a transit line, a water treatment upgrade, a bridge replacement).
- A specific facility (a landfill, a power plant, a hospital, a stadium) and its history and current condition.
- A public company's operations in one specific location.
- A local policy change and its documented effects.
- A university's or municipality's capital program.

Requirements: the topic must be documented in at least five categories, and it must be institutional rather than personal.

Instructions

Step 1 (15 min). Write the research question. One sentence, specific and answerable, with a named hypothetical consumer who would need the answer and what they would do with it. Then predict, before searching, which categories you expect to be most and least useful. You will compare this prediction against reality in Step 7.

Step 2 (100 min total). Research the topic in at least five of the following categories, spending roughly 20 minutes each. At least three must come from the official and compelled group.

Group	Categories
Official and compelled	Government records, court records, corporate filings, financial records, public databases
Published and edited	Books, newspapers, academic papers, radio, television, podcasts
Observational and technical	Satellite imagery, maps, metadata
Other	Social media, archived websites

For each category, complete a row of Worksheet 2A: what you searched, in which specific system, what you found, what you did not find, and the coverage boundary of your search.

Step 3 (20 min). Complete Worksheet 2B, the claim comparison matrix. Identify at least six discrete claims about your topic. For each, record which categories support it, which contradict it, and which are silent. Mark any claim supported by only one category.

Step 4 (15 min). Circularity check. For any claim supported by two or more items within the same category, determine whether those items are independent or share an origin. Report at least one instance of apparent corroboration that collapses on inspection, or state that you found none and describe how you checked.

Step 5 (10 min). Conflict resolution. Identify at least one point where categories disagree. Explain which you weighted more heavily and why, reasoning from production process rather than from preference.

Step 6 (25 min). Write the intelligence product. One page maximum, using the Chapter 1 template: BLUF with confidence and estimative language, rated key facts, an alternative explanation considered, two or more assumptions with consequences, two or more ranked gaps, and three recommendations. Every fact must indicate which category it came from.

Step 7 (15 min). Write the comparison analysis, one page. Which category delivered the most value per hour spent? Which delivered the least? How did that compare to your Step 1 prediction, and what does the difference teach you? Which category would you add if you had four more hours, and what would you expect it to settle?

Worksheet 2A: Category Research Record

Category	Specific system searched	Query or approach	Found	Not found	Coverage boundary	Value (H/M/L)	Time spent
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Worksheet 2B: Claim Comparison Matrix

Claim	Supporting categories	Contradicting categories	Silent categories	Single-category only?	Assessment
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Expected Outcomes

- Five or more categories genuinely researched, with at least three from the official and compelled group. Students who research five categories that are all media and web have not met the requirement.
- Specific systems named, not category labels. "County assessor parcel viewer, searched by address" scores 4; "public records" scores 1.
- Coverage boundaries stated for every negative result.

- Six or more discrete claims in the matrix, with at least one marked as single-category-only.
- At least one instance of within-category circularity detected, or a documented check showing none. In most topics, media coverage will show at least one instance.
- At least one genuine cross-category conflict, resolved by reasoning about production process.
- A comparison analysis where the prediction from Step 1 was at least partly wrong. This is the expected result and is rewarded, not penalized: most students overestimate media and social media and underestimate government records and archives.

Discussion Questions

1. Which category surprised you, in either direction? Why did you predict wrongly?
2. Your highest-value category cost you 20 minutes. What would you have found in two hours?
3. One claim rested on a single category. What would it take to corroborate it, and what would you write if you could not?
4. Where did categories conflict? Did you weight by process, or by which source felt more authoritative? Be honest about the difference.

Grading Rubric: Lab 2 (60 points)

Criterion	Weight	4 points looks like
Research question precision	x1	Specific, answerable, consumer named
Category breadth and group requirement	x3	Five-plus categories, three-plus official and compelled
Specific system naming	x3	Exact systems and approaches, never category labels
Coverage boundaries on negatives	x2	Present and precise for every negative finding
Claim comparison matrix	x3	Six-plus discrete claims, correctly attributed by category
Circularity detection	x2	Instance found and traced, or a documented check
Conflict resolution reasoning	x2	Weighted by production process, explicitly
Intelligence product quality	x3	BLUF, confidence, alternative, assumptions, ranked gaps, recommendations
Category attribution in the product	x1	Every fact indicates its category
Comparison analysis insight	x2	Prediction tested honestly;

Criterion	Weight	4 points looks like
		specific lesson drawn
Documentation standard	x2	Full collection log

Total possible: 96 weighted. Passing: 66.

Lab 3: Government and Court Records for a Fictional Case

Time required: 2 to 3 hours **Format:** Individual **Difficulty:** Intermediate

Objectives

1. Read a docket sheet and locate the filings that matter.
2. Distinguish allegations, exhibits, testimony, orders, and dispositions.
3. Cross-reference court records against government records to test claims.
4. Recognize name-match versus identity-match problems.
5. Produce a records-based findings memo with correct evidentiary characterization.

Prerequisites

- Sections 1.6 and 1.7.
- No internet access required; all material is provided.
- Completion of Labs 1 and 2.

The Requirement

Client: the risk committee of Harrow County, which is considering a five-year, \$11M contract with **Ridgeline Environmental Services LLC** for hazardous waste transport. **PIR:** Does the public record support Ridgeline's representations that it has (a) no adverse regulatory history, (b) no material litigation, and (c) operated its stated facility continuously for six years? **Deadline:** ten business days.

8.3.1 The Simulated Records Packet

All records below are fictional. Treat each as though you retrieved it from the stated system.

Record A: State corporate registry

Field	Value
Entity	Ridgeline Environmental Services LLC
Status	Active
Formed	6 years, 2 months ago

Field	Value
Registered agent	Corporate Agents of State X (commercial service)
Managers	R. Delacroix; M. Onyeka
Principal address	4820 Kiln Road, Harrow County

Record B: State corporate registry, second result

Field	Value
Entity	Ridgeline Environmental Services of the Northwest LLC
Status	Administratively dissolved, 3 years ago
Formed	9 years ago
Former name on file	Kiln Road Transport LLC
Managers	R. Delacroix
Principal address	4820 Kiln Road, Harrow County

Record C: County assessor, parcel record for 4820 Kiln Road

Field	Value
Owner of record	Kiln Road Holdings LLC
Acquired	9 years, 4 months ago
Land use code	I-2 (heavy industrial)
Improvements	Warehouse, 22,000 sq ft; office, 2,400 sq ft
Assessed value trend	Flat for 4 years, then a 31% increase 2 years ago

Record D: County permit history, 4820 Kiln Road

Permit	Date	Scope	Status
B-4471	8 years ago	Warehouse improvement, loading dock	Finaled
B-6620	5 years, 3 months ago	Containment berm installation	Expired, no final inspection
B-7702	2 years, 1 month ago	Warehouse expansion, 6,000 sq ft	Finaled
B-8115	11 months ago	Office remodel	Open

Record E: County code enforcement

Case	Date	Complaint	Resolution
CE-2231	5 years ago	Outdoor storage of drums beyond permitted area	Abated after 90 days
CE-3390	3 years, 6 months ago	Runoff to adjacent parcel	Referred to state environmental agency
CE-4102	14 months ago	Unpermitted structure	Resolved via permit B-8115

Record F: State environmental agency, enforcement database

Action	Respondent	Date	Finding	Penalty
NOV-1188	Kiln Road Transport LLC	7 years ago	Manifest recordkeeping deficiencies	\$4,200, paid
NOV-2074	Ridgeline Environmental Services of the Northwest LLC	4 years ago	Improper container labeling; secondary containment inadequate	\$18,500, paid
CO-0912	Ridgeline Environmental Services of the Northwest LLC	3 years, 8 months ago	Consent order requiring containment upgrades by a set date	Compliance status: closed, "administratively resolved upon entity dissolution"
(none)	Ridgeline Environmental Services LLC	(no actions)	(no actions)	(no actions)

Record G: State transportation agency, hazardous materials carrier registration

Field	Value
Carrier	Ridgeline Environmental Services LLC
Registration	Active, renewed annually for 6 years
Inspections	14 roadside inspections in 6 years; 3 with out-of-service violations (2 brakes, 1 placarding)
Safety rating	Satisfactory

Record H: Federal district court docket (partial)

Entry	Date	Filing
1	2 years, 3 months ago	Complaint, <i>Vance Aggregate Co. v. Ridgeline Environmental Services LLC</i> . Breach of contract and negligence; alleges improper disposal causing contamination of plaintiff's parcel. Damages sought: \$3.4M
4	2 years, 1 month ago	Answer; denies all substantive allegations; asserts comparative fault
17	1 year, 8 months ago	Plaintiff's motion for partial summary judgment, with Exhibits A through K
17-C	(same)	Exhibit C: waste manifests, 14 pages
17-F	(same)	Exhibit F: consultant's soil sampling report
22	1 year, 5 months ago	Order denying plaintiff's motion for partial summary judgment; court finds material facts in dispute
31	1 year, 1 month ago	Deposition excerpts of R. Delacroix filed in support of opposition
44	7 months ago	Stipulation of dismissal with prejudice following settlement; terms not filed

Record I: State trial court index, Harrow County (search results)

Case	Parties	Year	Type	Disposition
CV-19-0884	Delacroix, Robert v. Delacroix, Anne	7 years ago	Domestic	Closed
CV-21-2210	Sandoval Trucking v. Kiln Road Transport LLC	5 years ago	Contract	Judgment for plaintiff, \$61,400
CV-24-1177	Ridgeline Environmental Services LLC v. Meridian Insurance	1 year ago	Contract	Pending

Record J: County recorder, UCC and lien index

Instrument	Date	Party	Detail
UCC-7781	5 years ago	Equipment lessor	Six tank trailers
UCC-9043	2 years ago	Regional bank	All assets of Ridgeline Environmental Services LLC
Lien-3312	3 years, 9 months ago	State environmental agency	Recorded against 4820 Kiln Road; released 2 years, 10 months ago

Record K: Municipal check register, City of Harrow (public)

Date	Payee	Amount	Description
3 years ago	Ridgeline Environmental Services LLC	\$84,000	Household hazardous waste event
2 years ago	Ridgeline Environmental Services LLC	\$91,500	Household hazardous waste event
1 year ago	Ridgeline Environmental Services LLC	\$96,000	Household hazardous waste event

Steps

Step 1 (15 min). Restate the three representations as testable propositions. For each, write what evidence would confirm it and what would refute it.

Step 2 (20 min). Entity expansion. Complete Worksheet 3A from Records A, B, C, and the party names in H, I, and J. Draw the relationship structure. Note every shared address, shared manager, and former name.

Step 3 (25 min). Read the docket in Record H as an index. Complete Worksheet 3B: for each entry, state the document type and its evidentiary weight. Identify the entries you would retrieve first if you could retrieve only three, and justify the selection.

Step 4 (20 min). Test representation (a), adverse regulatory history. Complete Worksheet 3C using Records E, F, G, and J. State your finding and your confidence, and address the fact that Record F shows no actions against the contracting entity itself.

Step 5 (20 min). Test representation (b), material litigation, using Records H and I. Apply the allegation-versus-finding distinction rigorously and address the name-match problem in Record I.

Step 6 (15 min). Test representation (c), six years of continuous operation, using Records A through D

and K.

Step 7 (30 min). Write the findings memo. Two pages maximum. Required: BLUF with confidence; findings organized by representation, with facts separated from assessments; the entity structure stated without alleging intent; three or more assumptions; four or more ranked gaps including what you would request that is not in this packet; and five or more recommendations to the risk committee.

Step 8 (10 min). Write the characterization check: list every sentence in your memo that could be misread as asserting wrongdoing, and show the wording you used instead.

Worksheet 3A: Entity Expansion

Entity	Former names	Status	Managers	Address	Relationship
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Worksheet 3B: Docket Analysis

Entry	Document type	Allegation / evidence / finding / disposition	Evidentiary weight	Retrieve first?
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Worksheet 3C: Representation Testing

Representation	Supporting records	Contradicting records	Determination	Confidence	Coverage boundary
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Expected Outcomes

- **Entity structure.** Three related entities sharing one address and one manager, with the dissolved entity's former name (Kiln Road Transport LLC) tying Record F's oldest violation into the family. The parcel is held by a fourth entity, Kiln Road Holdings LLC. A student who reads only the contracting entity's own records finds almost nothing.
- **Representation (a) is not supported.** The contracting entity has a clean enforcement record, but two related entities under the same manager at the same address accumulated a notice of violation, a second notice with a penalty, a consent order, and a recorded state lien on the property. The consent order closed "administratively resolved upon entity dissolution," which means the compliance obligation was never demonstrated as completed. Permit B-6620 for the containment berm expired without final inspection, which corroborates the same gap from a different system. Strong students connect the consent order, the expired permit, and the runoff complaint into one finding.
- **Representation (b) is not supported as to materiality.** A \$3.4M federal case existed, and the correct characterization is precise: the complaint contained allegations, the court denied summary judgment because material facts were in dispute (which is not a finding for either party), and the case ended in a stipulated dismissal following settlement with terms not filed,

which establishes no facts. There is also a pending case in which Ridgeline is the plaintiff against its insurer, which is itself informative about the settlement.

- **Name match trap.** Record I's first case is a domestic matter involving individuals sharing a manager's surname. The correct handling is to exclude it entirely as irrelevant and as involving private individuals, and to say so in the ethics note rather than silently dropping it.
- **Representation (c) is partially supported.** The contracting entity has been registered and carrier-registered for six years and has three years of municipal payments. Continuous operation *of the facility* spans nine years across predecessor entities, which is a different claim from six years of operation by this entity. Precise students distinguish the two.
- **Gaps** should include the settlement terms (not filed), the exhibits themselves (identified but not retrieved), the soil sampling report, state courts outside Harrow County, other states where the entities may have operated, and the insurance dispute's subject matter.
- **Characterization.** No sentence should assert that Ridgeline caused contamination or violated any law. The defensible framing throughout is that the record shows specified filings, findings, and gaps.

Discussion Questions

1. The contracting entity's own enforcement record is clean. Is the representation therefore technically accurate? Answer as an analyst, then as the risk committee.
2. The consent order closed on dissolution. Explain to a non-lawyer why that is a finding worth flagging.
3. Which three docket entries would you retrieve, and what would each settle that the docket sheet alone cannot?
4. Record I contains a domestic case. Explain your handling and the rule you applied.
5. A settlement establishes no facts. How do you describe a \$3.4M settled case in a memo without implying liability?

Grading Rubric: Lab 3 (64 points)

Criterion	Weight	4 points looks like
Propositions restated with refutation criteria	x2	All three, both directions specified
Entity expansion completeness	x3	Four entities mapped, former name thread used
Docket type and weight analysis	x3	Every entry correctly typed; weights accurate
Retrieval prioritization	x2	Three chosen with settling-value justification

Criterion	Weight	4 points looks like
Regulatory finding	x3	Related-entity history connected across three systems
Consent order and expired permit connection	x2	Explicitly linked as a single compliance gap
Litigation characterization precision	x3	Allegation, denial of summary judgment, and settlement each correctly described
Name-match handling	x2	Domestic case excluded, rule stated, ethics note present
Continuous operation distinction	x2	Entity tenure distinguished from facility tenure
Assumptions and ranked gaps	x2	Three-plus assumptions, four-plus gaps, ranked
Recommendations	x2	Five-plus, specific and assignable
Characterization check	x2	No sentence implies wrongdoing; alternates shown
Coverage boundaries	x2	Systems searched stated explicitly

Total possible: 120 weighted. Passing: 82.

Lab 4: Comparing Archived Versions of a Website

Time required: 2 to 2.5 hours **Format:** Individual **Difficulty:** Intermediate

Objectives

1. Locate and enumerate archived captures of a site.
2. Compare captures systematically against defined elements.
3. Date changes to the precision the capture interval allows, and no further.
4. Distinguish substantive change from cosmetic change.
5. Produce a change chronology with correct dating language.

Prerequisites

- Section 1.17.
- Internet access.

- Completion of Labs 1 through 3.

Subject Selection

Choose an **organization's website**. Not an individual's. Requirements: the organization has existed for at least eight years, the site has at least twelve archived captures spanning at least six years, and the organization is institutional (a company, agency, university, nonprofit, trade association, or municipality).

Good candidates: a mid-sized company's site, a municipal department, a trade association, a professional services firm, a nonprofit. Sites that change constantly (large news outlets) are poor choices, because change volume swamps signal.

Steps

Step 1 (15 min). Enumerate captures. Complete Worksheet 4A: list every capture date available for the site's main page, and note the capture frequency pattern over time. Comment on what the pattern itself suggests (a site captured monthly is popular or heavily linked; one captured twice in five years is obscure).

Step 2 (10 min). Select six comparison points with a stated purpose: earliest available, latest available, and four intermediate captures chosen to bracket periods of interest or to sample evenly. State your selection rationale.

Step 3 (60 min). Systematic comparison. For each of the six captures, complete Worksheet 4B across all of these elements:

Element	What to record
Page title and tagline	Exact text
Navigation menu items	Full list
Headline claim	The main assertion on the page
Numeric claims	Any figures: years in business, clients served, locations, employees, capacity
Named clients, partners, or funders	Full list where present
Named leadership	Names and titles where present
Locations or addresses	All listed
Services or products offered	Full list
Certifications, awards, memberships	All claimed
Terms, policies, disclaimers	Presence and any substantive change
Contact details	Phone, email, physical address
Linked documents	Any PDFs or files, and whether they captured
Missing elements in the capture	Images, scripts, or subpages that failed to archive

Step 4 (20 min). Build the change chronology on Worksheet 4C. For each substantive change: what changed, the capture date it was last seen in its prior state, the capture date it was first seen changed, and therefore the bracketed window. Use the correct language: "changed between [date] and [date]," never a specific date you cannot support.

Step 5 (10 min). Classify each change as substantive or cosmetic, with a one-line reason. Redesigns, font changes, and reorganized menus are cosmetic unless they remove content.

Step 6 (15 min). Corroboration. Select the two most significant substantive changes and attempt to corroborate each from a different category: a news item, a filing, a registry record, a permit, a job posting, or a press release. Record what you found, including failures.

Step 7 (20 min). Write the findings memo. One page: BLUF with confidence, a change chronology table, the two corroborated changes with their supporting category, at least two assumptions, at least two gaps including what the capture intervals prevent you from knowing, and an assessment of what the change pattern suggests about the organization. Cite capture URLs.

Step 8 (10 min). Limits statement. Half a page: what the archive did not capture, what the capture intervals prevent you from dating, and one specific finding you suspect but cannot support.

Worksheet 4A: Capture Enumeration

Year	Number of captures	Notable gaps	Pattern observation
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Worksheet 4B: Element Comparison

Element	Capture 1 (date)	Capture 2	Capture 3	Capture 4	Capture 5	Capture 6
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Worksheet 4C: Change Chronology

Change	Last seen prior state (date)	First seen changed (date)	Bracketed window	Substantive or cosmetic	Corroboration attempted
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Expected Outcomes

- Twelve or more captures enumerated with a pattern observation.
- Six captures compared across all thirteen elements, with exact text recorded rather than paraphrase.
- At least four substantive changes identified. In most organizational sites over six years, expect changes to numeric claims, leadership names, client or partner lists, and service offerings.
- Dating language correct throughout. Any student who writes "the claim changed in March" without two bracketing captures on either side of March loses points on the criterion that

matters most in this lab.

- At least one element that failed to capture, honestly reported.
- At least one corroboration attempt that failed, reported as a finding.
- A limits statement that names a suspected but unsupportable finding. This is the mark of an analyst who understands the boundary between what the evidence shows and what they believe.

Discussion Questions

1. What did the capture frequency pattern tell you before you compared anything?
2. Which changed element was most informative, and why was it more informative than the others?
3. You bracketed a change to a window of several months. What other category could narrow it, and to what precision?
4. Organizations remove claims for many innocuous reasons. How did you avoid over-reading a removal?
5. What did you suspect but could not support? What would resolve it?

Grading Rubric: Lab 4 (56 points)

Criterion	Weight	4 points looks like
Subject selection compliance	x1	Institutional, sufficient capture history, appropriate change volume
Capture enumeration and pattern observation	x2	Twelve-plus captures, pattern interpreted
Comparison point selection rationale	x1	Purposeful, stated
Element comparison completeness	x3	All thirteen elements across six captures, exact text
Substantive change identification	x3	Four-plus real changes found
Dating language precision	x3	Bracketed windows only; no unsupported specific dates
Substantive versus cosmetic classification	x2	Correct, with reasons
Cross-category corroboration	x3	Two attempted from other categories, failures reported
Missing-capture honesty	x2	Failures to archive reported, not glossed
Findings memo quality	x2	BLUF, confidence, chronology,

Criterion	Weight	4 points looks like
		assumptions, gaps
Limits statement	x2	Names a suspected but unsupportable finding

Total possible: 96 weighted. Passing: 66.

Lab 5: Metadata Examination

Time required: 2 to 2.5 hours **Format:** Individual **Difficulty:** Intermediate to advanced

Objectives

1. Extract metadata from several file types.
2. Distinguish what metadata establishes from what it merely suggests.
3. Use metadata to test a claim about a document's provenance.
4. Identify metadata that constitutes an organizational exposure.
5. Explain, in writing, both how metadata can support and how it can refute an investigation.

Prerequisites

- Section 1.19.
- The sample file set provided with this workbook.
- No specialized software required; the extraction methods below use widely available tools, and a printed metadata listing is provided as a fallback in Section 8.5.3 so the lab can be completed without any tooling.

The Scenario

You support a county procurement office. A vendor, **Calder Fabrication LLC**, has submitted a bid package. Separately, an anonymous complaint has arrived, attaching a document alleged to be Calder's internal cost sheet and claiming it proves bid rigging. The procurement officer needs to know what the files themselves reveal.

Timeline anchors for this scenario. The bid deadline was **30 April 2026**. The anonymous complaint arrived on **14 May 2026**. The relative offsets used in the metadata listing in Section 8.5.3 are measured from those two dates, and the timestamps in the sample files themselves use them.

You have five sample files:

File	Alleged origin
calder-bid-cover.pdf	Calder's official bid cover letter

File	Alleged origin
calder-capability-statement.docx	Calder's capability statement, submitted with the bid
site-photo-1.jpg	Photograph Calder submitted showing its fabrication floor
site-photo-2.jpg	Second photograph, same submission
internal-cost-sheet.pdf	The document attached to the anonymous complaint

8.5.1 Extraction Methods

Any one of these is sufficient. Use whichever is available to you.

Method 1: file properties in your operating system. Right-click, then Properties or Get Info, then the Details or More Info panel. Shows a useful subset: author, dates, camera information, and dimensions.

Method 2: the application itself. Open a PDF in a reader and look at Document Properties. Open a word processing document and look at File, then Info or Properties. Both expose author, creating application, timestamps, revision counts, and editing time.

Method 3: a metadata viewer. Command-line and graphical metadata tools read a much larger field set, including GPS coordinates, software history, and embedded thumbnails. If you have one available, use it, and report which tool and version you used.

Method 4: the provided listing. Section 8.5.3 contains the extracted metadata for all five files. Use it if you cannot run any of the above, and note in your submission that you used the listing.

Whichever method you use, **record the method in your collection log**, because a limited extraction method producing fewer fields is a coverage boundary like any other.

Steps

Step 1 (10 min). Before extracting anything, write down what you expect to find and what each file would need to show to be consistent with its alleged origin. Predicting first prevents you from reading whatever you find as confirmation.

Step 2 (35 min). Extract metadata from all five files. Complete Worksheet 5A: one row per field per file. Record every field your method exposes, including fields that are empty, since an empty field is information.

Step 3 (20 min). Internal consistency analysis. Complete Worksheet 5B. For each file, assess whether the metadata is internally consistent: do creation and modification timestamps make sense in sequence, does the creating application match the claimed author's environment, do the timestamps fit the bid timeline, is the device consistent across files claimed to come from the same source?

Step 4 (20 min). Cross-file analysis. Complete Worksheet 5C. Which files share an author, a creating application, a device, or a template? What relationships does that establish, and what does it merely

suggest?

Step 5 (20 min). The provenance test. Focus on `internal-cost-sheet.pdf`. Assess whether its metadata supports, contradicts, or is silent on the claim that it is Calder's internal document. State a confidence level. Then, critically, list at least four alternative explanations for the metadata pattern you observe that do not involve fabrication, and explain how each would change the assessment.

Step 6 (15 min). The exposure audit. Identify every field across all five files that would constitute an unintended disclosure if the files were published. For each: what it reveals, and the control that would have prevented it.

Step 7 (25 min). Write the memo. Two pages maximum. Required sections: BLUF with confidence; findings per file with facts separated from inference; the provenance assessment with alternatives; the exposure audit; at least three assumptions; at least three gaps; and a closing section titled "**How metadata could support and how it could refute this investigation,**" in which you write one paragraph on each. That final section is worth the most points in the rubric.

8.5.2 Worksheets

Worksheet 5A: Extracted Metadata

File	Field	Value	Extraction method
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Worksheet 5B: Internal Consistency

File	Timestamp sequence sensible?	Application consistent with claimed origin?	Fits the bid timeline?	Anomalies
------	------------------------------	---	------------------------	-----------

Worksheet 5C: Cross-File Relationships

Shared attribute	Files sharing it	What it establishes	What it only suggests
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Worksheet 5D: Exposure Audit

File	Field	What it discloses	Preventive control
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8.5.3 Provided Metadata Listing (fallback)

Use this only if you cannot extract metadata yourself. The values below are what a full extraction returns for the sample files.

calder-bid-cover.pdf

Field	Value
Title	Calder Fabrication Bid Cover Letter
Author	J. Marchetti
Subject	RFP 24-118 Response
Creating application	Enterprise word processing suite, version consistent with a business installation
PDF producer	Same suite's PDF export module
Creation date	14 days before the bid deadline, 09:12
Modification date	14 days before the bid deadline, 09:31
Keywords	(empty)

calder-capability-statement.docx

Field	Value
Author	J. Marchetti
Last modified by	J. Marchetti
Company	Calder Fabrication LLC
Creation date	21 days before the bid deadline, 15:40
Modification date	14 days before the bid deadline, 08:58
Revision number	17
Total editing time	4 hours 22 minutes
Template	CalderLetterhead.dotx
Comments present	Yes: two reviewer comments remain in the file
Tracked changes present	Yes: one deletion of a sentence describing a prior project

site-photo-1.jpg

Field	Value
Camera make and model	Consumer DSLR, mid-range model
Lens	18-55mm zoom
Original capture date	26 days before the bid deadline, 11:03
Modification date	15 days before the bid deadline, 17:22
Software	Photo editing suite, current version
GPS coordinates	Present; resolve to the address Calder lists as its facility

Field	Value
Orientation	Rotated
Embedded thumbnail	Present; shows a wider frame than the visible image, including a second building

site-photo-2.jpg

Field	Value
Camera make and model	Smartphone, different manufacturer from photo 1
Original capture date	3 years, 2 months before the bid deadline, 14:47
Modification date	15 days before the bid deadline, 17:25
Software	Same photo editing suite as photo 1
GPS coordinates	Absent
Embedded thumbnail	Present; matches the visible image

internal-cost-sheet.pdf

Field	Value
Title	(empty)
Author	(empty)
Creating application	Consumer spreadsheet application, free tier
PDF producer	Generic PDF printer driver
Creation date	2 days before the anonymous complaint arrived, 22:41
Modification date	2 days before the complaint arrived, 22:44
Keywords	(empty)
Fonts embedded	Two default system fonts only
Layer or form structure	None

Expected Outcomes

- **The bid documents are internally consistent** with their claimed origin: matching author, a corporate template, a company field naming the vendor, a plausible 17-revision and four-hour editing history, and timestamps that fit the bid timeline. Confidence that these originated inside Calder: high.
- **The capability statement contains two exposures:** residual reviewer comments and a tracked deletion of a sentence about a prior project. Strong students recognize that the deleted sentence

is itself a lead worth pursuing through other categories, and that the vendor's document control process failed.

- **Photo 1 has a GPS fix matching the claimed facility**, which supports the submission, and an **embedded thumbnail showing a wider frame including a second building**. That is the most interesting finding in the set: the original image contained more than the submitted crop. Its significance is unresolved on this evidence alone and should be flagged as a lead, not a conclusion.
- **Photo 2 predates the bid by more than three years**. The correct finding is that a photograph presented as showing current facilities was captured three years earlier. That is a factual inconsistency worth reporting, and it is not proof of deception: reuse of older photography is extremely common and may be entirely innocent. The absence of GPS in photo 2 is uninformative, because many devices and settings do not record it.
- **Both photos were edited with the same software within three minutes of each other**, fifteen days before the deadline. That establishes common processing, consistent with routine preparation of a bid package.
- **The cost sheet's metadata does not support its claimed origin**. No author, no title, a free-tier consumer spreadsheet application rather than the enterprise suite evidenced in the vendor's own documents, a generic print-to-PDF producer, only default fonts, no template, and creation three minutes before modification two days before the complaint arrived. Assessment: the claimed provenance is not supported, at moderate confidence.
- **The four alternative explanations are the highest-value part of Step 5**. Acceptable alternatives include: a genuine document re-created or retyped by a recipient to protect a source; a genuine document printed and re-scanned or re-exported, which resets all fields; a genuine document exported from an internal system that produces generic metadata; a genuine document opened and re-saved by an intermediary; or deliberate metadata scrubbing by a source protecting themselves. A student who concludes "fabricated" without engaging these has failed the central lesson, and the rubric weights it accordingly.
- **Exposure audit** should identify: author name in three files, company field, template filename revealing internal naming, residual comments, tracked deletion, GPS coordinates, and the embedded thumbnail revealing pre-crop content. Controls: metadata stripping in the publication process, comment and tracked-change removal, image re-export rather than crop-in-place, and a pre-release checklist.
- **The final section** should distinguish clearly: metadata supports an investigation by establishing time, device, authorship, relationships, and internal consistency, and by surfacing content the publisher did not intend to release; metadata refutes an investigation by contradicting a claimed provenance, by exposing that an apparent smoking gun has an implausible origin, and by demonstrating that a document a party relied on was created later than claimed. Both directions must be present.

Discussion Questions

1. Photo 2 is three years old. Write the sentence you would put in a memo, and then write the sentence a careless analyst would write. What is the difference?
2. The embedded thumbnail showed more than the submitted image. Is that a finding, a lead, or nothing? Defend your answer.
3. You listed four innocent explanations for the cost sheet's metadata. Which is most likely, and what single piece of additional evidence would distinguish among them?
4. The capability statement retained a tracked deletion. How would you pursue the deleted content through other categories, and what would you not do?
5. If a document's metadata had been perfectly consistent with its claimed origin, would that prove authenticity? Explain in two sentences.

Grading Rubric: Lab 5 (68 points)

Criterion	Weight	4 points looks like
Prediction before extraction	x1	Written first, specific, and later compared honestly
Extraction completeness	x2	All five files, all available fields, empty fields recorded
Extraction method documented	x1	Method and tool named as a coverage boundary
Internal consistency analysis	x3	Timestamp sequences, application fit, and timeline fit assessed per file
Cross-file relationship analysis	x2	Shared attributes identified; establishes versus suggests distinguished
Photo 2 date finding	x2	Reported precisely as an inconsistency, not as deception
Embedded thumbnail finding	x2	Identified and correctly characterized as a lead
Provenance assessment	x3	Correct direction, calibrated confidence, reasoning stated
Alternative explanations	x4	Four-plus genuine innocent alternatives, each with its effect on the assessment
Exposure audit	x2	Seven-plus exposures with preventive controls
Fact / inference separation	x2	Visible throughout

Criterion	Weight	4 points looks like
Support-and-refute section	x4	Both directions developed substantively, with specifics from this exercise
Assumptions and gaps	x2	Three-plus of each
Length discipline	x1	Two pages

Total possible: 124 weighted. Passing: 86.

Part 9: Capstone Assignment

The Ambervale Renewables Investigation

Weight: 30% of chapter grade **Time required:** 5 to 8 hours **Format:** Individual, may be completed across multiple sittings **Deliverables:** Six items, listed in Section 9.4

9.1 The Scenario

You have been engaged by the **Tolliver County Planning Commission**, a fictional body, as an open source analyst. Everything in this scenario is fictional.

The decision. In seven weeks the commission votes on a conditional use permit for a 40-acre battery energy storage facility proposed by **Ambervale Renewables LLC** at 1170 Quarry Line Road. The application represents that Ambervale:

1. Has developed and currently operates three comparable facilities in other states.
2. Is financially capable of completing the \$62M project and funding decommissioning.
3. Has no adverse environmental or safety record.
4. Has secured the site, with no title or access impediments.
5. Has community support, evidenced by a petition with 900 signatures.

Complications. A local group opposes the project and has made its own claims: that Ambervale is a shell for an out-of-state parent with a poor record, that the site is in a floodplain, that the access road cannot carry construction traffic, and that the support petition is fabricated. The commission wants an evidence-based assessment of all of it, and it wants to be able to defend whatever it decides.

Your position. You report to the commission's staff director. Five business days. No budget beyond ordinary public records fees.

9.2 The Provided Reporting

Thirty items. Some are decisive, some are noise, at least two conflict, at least one is probably false, and at least one raises an ethical question you must handle explicitly.

#	Category	Source	Content
1	Corporate filings	State X registry	Ambervale Renewables LLC. Active. Formed 22 months ago. Registered agent: commercial service. One manager: Ambervale Holdings LLC
2	Corporate filings	State Y registry	Ambervale Holdings LLC. Active. Formed 6 years ago. Managers: two named individuals. Principal address in State Y
3	Corporate filings	State Y registry	Three additional entities named "Ambervale [Project Name] LLC," each formed within the past 5 years, each listing Ambervale Holdings LLC as manager
4	Corporate filings	State Z registry	"Coldbrook Energy Partners LLC," dissolved 3 years ago. Former name on file: "Ambervale Energy LLC." One manager name matches an Ambervale Holdings manager
5	Archived websites	Capture of Ambervale's site, 4 years ago	Listed four projects, including one described as "operating" in State Z
6	Internet	Ambervale's current site	Lists three projects, all described as "operating." The State Z project no longer appears
7	Public databases	Federal energy facility database	Two of the three claimed projects appear with operating status.

#	Category	Source	Content
			The third appears with status "under construction"
8	Newspapers	Trade press, 3 years ago	Article describing a battery facility fire in State Z at a project developed by "Ambervale Energy," with a nine-day emergency response
9	Government records	State Z fire marshal report (public)	Findings on the same incident: thermal runaway in one enclosure; contributing factors included inadequate spacing and a deficient emergency response plan. Respondent named: Ambervale Energy LLC
10	Court records	Federal district court, State Z	Case filed 3 years ago against Ambervale Energy LLC by an adjacent property owner. Complaint alleges property damage from the fire. Docket shows stipulated dismissal following settlement 18 months ago; terms not filed
11	Court records	State trial court, State Y	Pending case: Ambervale Holdings LLC v. a construction contractor. Contract dispute, \$4.1M claimed
12	Maps	County GIS parcel viewer	1170 Quarry Line Road is two parcels totaling 41.2 acres, not one 40-acre parcel
13	Government records	County assessor	Parcel A (28.6 acres) owner of record: a family trust. Parcel B (12.6 acres) owner of

#	Category	Source	Content
			record: Quarry Line Holdings LLC
14	Government records	County recorder	An option agreement memorandum recorded 14 months ago covering Parcel A. No recorded instrument covering Parcel B
15	Corporate filings	State registry	Quarry Line Holdings LLC. Active. Manager: a local named individual, unaffiliated with Ambervale on the record
16	Maps	Federal flood hazard mapping	Approximately 6 acres of Parcel B lie within the mapped 100-year floodplain. The remainder of both parcels lies outside it
17	Government records	County road department, published pavement inventory	Quarry Line Road: 22-foot width, last overlay 19 years ago, rated "poor." No posted weight restriction
18	Government records	County road department meeting minutes, 8 months ago	Staff report identifying Quarry Line Road as a candidate for reconstruction, unfunded
19	Satellite imagery	Commercial imagery, current	Both parcels appear as cleared agricultural land. A gravel access track connects Parcel A to Quarry Line Road
20	Satellite imagery	Imagery from 4 years ago	Same land use. A structure present on Parcel B in the older image is absent in the current image
21	Public databases	State environmental agency	No enforcement actions against Ambervale Renewables LLC or

#	Category	Source	Content
			Ambervale Holdings LLC in this state
22	Public databases	State Z environmental agency	One notice of violation against Ambervale Energy LLC, 4 years ago, relating to stormwater controls during construction. Penalty paid
23	Financial records	Municipal bond continuing disclosure, unrelated jurisdiction	A different jurisdiction's disclosure references a payment default by "Ambervale Energy LLC" on a lease obligation 4 years ago
24	Corporate filings	State Y registry, annual reports	Ambervale Holdings LLC annual reports filed on time each year. No financial information required or included
25	Metadata	The application PDF submitted to the county	Author field: a named individual. Creating application: an enterprise suite. Creation date: 9 days before submission. Contains a tracked deletion of a sentence beginning "Our State Z project experienced"
26	Metadata	The petition PDF submitted with the application	Author field: empty. Creating application: a free online form service export. Creation date: 3 days before submission. 900 rows
27	Social media	Local community group page	60 posts opposing the project over 6 weeks, from a mix of long-standing local accounts and four accounts created in the past month posting identical text

#	Category	Source	Content
28	Newspapers	Local paper, 2 weeks ago	Article covering the controversy, quoting a named opposition organizer and an Ambervale spokesperson. Sourcing for the "shell company" claim is attributed to "opponents"
29	Podcasts	Industry podcast, 7 months ago	An Ambervale Holdings executive discusses the company's pipeline, states that "two of our three operating assets are fully energized," and describes a financing structure dependent on a tax credit transfer closing in the current year
30	Public databases	Regional commodity report	Steel and battery cell input prices rose 8% over the past two quarters

9.3 Guidance on the Substance

A weak submission produces five lists, one per representation, and leaves the commission to weigh them. A strong submission notices which representations are cleanly resolvable from the record, which are partly true in a way that matters, and which cannot be resolved from open sources at all, and says so in that order.

You will also find that the opposition's claims are not uniformly true or false either. Assessing each side's claims by the same standard is the professional obligation here, and a submission that is rigorous toward one side and credulous toward the other will score poorly regardless of which side it favors.

Items 25 and 26 are metadata. Item 27 concerns accounts. One item in the packet concerns an identifiable private individual in a way you must handle explicitly.

9.4 What You Must Produce

Deliverable 1: Source Plan With Justification (15 points)

Written as though on day one, before reading the packet closely.

- **Five PIRs**, one per representation, each tied to the commission's decision and its deadline.
- **For each PIR: three to five EEIs**, and for each EEI, two to four observable indicators.
- **A source plan table**: for each indicator, the category, the specific source system you would query, the reason that category is the right one (this is the justification requirement, and it is graded), the access model, the expected latency, and the coverage boundary you expect.
- **At least three questions that open sources cannot answer**, identified honestly, with the non-open source or process that would answer each.
- **A prohibited-activity statement**: at least five specific things you will not do on this engagement.

Deliverable 2: Classification and Evaluation (15 points)

All 30 items in a table with: category; primary, secondary, or tertiary; structured, semi-structured, or unstructured; reliability (A-F) and credibility (1-6) with one-line justifications; which representation or opposition claim it bears on; and relevance (relevant, possibly, or noise).

Then, in one paragraph each: the two items that conflict and how you resolved the conflict; the item you assess as probably false and why; and the item raising an ethical question and how you handled it.

Deliverable 3: Corroboration Matrix (20 points)

For each of the five applicant representations and each of the four opposition claims, a matrix row showing:

- The claim, stated precisely.
- Every item supporting it, by category.
- Every item contradicting it, by category.
- **The number of distinct categories on each side**, since this is the measure that matters.
- Whether any apparent corroboration collapses on inspection.
- Determination: supported, partly supported, not supported, or unresolved.
- Confidence level and the reason for it.

A determination resting on a single category must be labeled as such.

Deliverable 4: Reliability Analysis (10 points)

A short analysis, one page maximum, addressing:

- Which two items in the packet carry the highest reliability and why, reasoning from production process.
- Which two carry the lowest and why.
- One item where a reliable source carries a low-credibility claim, or the reverse, with your handling.
- One instance of within-category corroboration that is weaker than it appears.
- How you weighted the applicant's own documents against compelled records.

Deliverable 5: Professional Intelligence Report (30 points)

Maximum three pages, plus the collection log and ethics annex, which do not count toward the limit. Addressed to the staff director, written in the knowledge that it will be read into the public record and quoted by both sides.

Required structure:

1. **Bottom line.** Four sentences maximum, with a confidence level and estimative language. Must distinguish what is resolved from what is unresolved.
2. **Applicant representations.** Each of the five, with facts (rated, with category noted) separated from assessments.
3. **Opposition claims.** Each of the four, assessed by the same standard.
4. **Site and infrastructure findings.** Parcels, title status, floodplain, and road capacity.
5. **Entity structure.** The Ambervale, Ambervale Holdings, Coldbrook, and project-entity relationships, stated without alleging intent.
6. **Key assumptions.** At least four, each with what changes if it is wrong.
7. **Intelligence gaps.** At least five, ranked by how much resolving each would change the assessment, and each identifying the source that would resolve it.
8. **Recommended actions.** At least seven, specific, assignable, and time-bound. At least two executable this week at no cost. At least two must be conditions or information requests the commission could impose or make rather than conclusions.
9. **What would change this assessment.** Two to four sentences.
10. **Collection log** meeting the documentation standard for at least twenty items, including the method used for metadata extraction.
11. **Ethics annex**, one page maximum: every individual appearing anywhere in the work and how each was handled, what you deliberately did not collect, and your specific handling of the private individual in the packet and of the accounts in item 27.

Deliverable 6: Rationale Annex (10 points)

One page maximum:

- Why you chose your bottom-line judgment out of everything available.
 - Why you chose your confidence level and not the level above or below.
 - Which three items you excluded that a less careful analyst would have included, and why.
 - Which finding you were most tempted to overstate, and the wording you used instead.
 - One sentence on how you kept the same standard for both sides.
-

9.5 Capstone Grading Rubric (100 points)

Deliverable	Criterion	Points
1. Source plan	Five PIRs, decision-linked and answerable	4
	EEIs sufficient; indicators observable	3
	Category justification for each source (the graded element)	4
	Coverage boundaries anticipated	2
	Non-open-source questions identified	1
	Prohibited-activity statement	1
2. Classification	Category, type, and structure accuracy	4
	Reliability and credibility ratings defensible	4
	Relevance and noise identification	3
	Conflict resolution reasoned	2
	Probable falsehood identified	1
	Ethical item identified and handled	1
3. Corroboration matrix	All nine claims stated precisely	3
	Support and contradiction correctly attributed by category	5
	Distinct-category counts present and correct	4

Deliverable	Criterion	Points
	Collapsing corroboration identified	3
	Determinations defensible, single-category findings labeled	3
	Confidence levels with stated reasons	2
4. Reliability analysis	Highest and lowest reliability, reasoned from process	4
	Reliable-source, low-credibility case handled	3
	Weighting of applicant documents against compelled records	3
5. Report	Bottom line: four sentences, confidence, resolved versus unresolved	4
	Applicant representations assessed with facts separated	4
	Opposition claims assessed to the same standard	4
	Site and infrastructure findings precise	3
	Entity structure without allegation of intent	3
	Assumptions with consequences	2
	Gaps ranked, each with its resolving source	3
	Recommendations, including no-cost and condition-based options	3
	Collection log: twenty-plus items at full standard	2
	Ethics annex	2
6. Rationale	Confidence defended against adjacent levels	3
	Exclusions justified	3
	Overstatement temptation identified with the wording used	2
	Even-handedness statement supported by the report itself	2

Deliverable	Criterion	Points
Total		100

Grade bands: 90 to 100 exemplary, 80 to 89 proficient, 70 to 79 competent, 60 to 69 developing, below 60 not yet passing.

Automatic deductions:

- Any judgment without a confidence level: minus 5 per instance.
- Any finding not traceable to a logged source: minus 3 per instance.
- Treating a complaint's allegations or a settlement as established fact: minus 8 per instance.
- Asserting a specific date for a change that the evidence brackets only to a window: minus 4 per instance.
- No coverage boundary on a negative finding: minus 3 per instance.
- Collecting or reporting anything about the private individual in item 15 beyond their appearance as a manager of record on a public filing: minus 25. The correct handling is to treat the name as an entity-relationship data point, note in the ethics annex that no further research on that individual was conducted, and state that the absence of a recorded relationship to Ambervale is a gap rather than a finding about the person.

Part 10: Answer Keys

Instructors may wish to remove this section from student copies.

10.1 Short Answer Key

1. A category is a class of information defined by its producer, process, and obligations; a tool is a means of searching one or more categories. Confusing them causes analysts to treat search results as the universe of available information, which systematically excludes the deep web where most authoritative material lives.

2. Content, production process, access model, coverage, latency, and failure mode. Production process determines reliability, because obligations, defined formats, review, and penalties are what make a record dependable, rather than the medium or the publisher's familiarity.

3. Primary is an original first-hand record (a recorded deed). Secondary describes or interprets it (a news article about the transaction). Tertiary compiles secondary material (an encyclopedia entry). The rule is to walk every claim you rely on back to its primary source.

4. Primary means closer to the event, not accurate. A police report is a primary record of one participant's account. A corporate filing is a primary document written to satisfy a regulator and to present the filer favorably within the rules. Both are primary and both are interested.

5. Structured data has defined fields with consistent types; unstructured is free-form. Converting an unstructured question into a structured one produces a bounded, reproducible answer with an explicit coverage statement, whereas keyword searching produces ranked results whose completeness cannot be assessed.
6. The codebook rule is that a field's definition must be read before the field is used. Example: "status: active" in a corporate registry may mean only that the entity has not been administratively dissolved, not that it conducts business; or "inspection date" may be the report date rather than the visit date.
7. Surface web is crawled and indexed. Deep web is on the ordinary internet but unindexed, behind queries, forms, logins, or paywalls. Dark web requires overlay network software. The common confusion is treating "deep web" and "dark web" as synonyms, when the deep web is where court records, registries, and library catalogs live.
8. Because search engines index only the crawlable surface, rank results by relevance models rather than completeness, and personalize by user and time. The authoritative material must be reached by going to the system that holds it and querying it.
9. Depth on a bounded subject, coverage of periods before digitization, and curated endnote trails that identify primary sources someone else has already located.
10. Because statutory publication requirements compel disclosure of foreclosures, probate, name changes, trade names, procurement solicitations, tax delinquency, zoning hearings, license applications, and dissolutions, in a searchable medium, frequently in a form available nowhere else to a member of the public.
11. Whether the reporter was present (byline and dateline); what the stated sourcing is and whether it is checkable; whether the piece is original or a wire or aggregated rewrite; and whether the outlet has a visible correction practice.
12. Whether it was peer reviewed and where; the size and relevance of the sample; whether it is a single study or a systematic review; how it has been cited and by whom; funding and declared conflicts; and whether it has been retracted.
13. Because a missing record may mean the jurisdiction does not require it, has not digitized it, files it under a different name, or lies outside the system's scope. Example: a registry search returning nothing means the entity is not registered in that state under that exact name, not that it does not exist.
14. The docket is the index; the complaint contains one party's allegations; the answer contains denials; motions contain arguments; exhibits contain documents produced in discovery; orders and opinions contain the court's findings; the judgment or disposition is the outcome. A complaint alleges, an order finds, and a settlement resolves without establishing facts.
15. Because discovery compels production of internal material never intended for publication, and exhibits attached to filings become public. Internal memoranda, contracts, studies, and correspondence enter the record as exhibits.
16. Radio functions when power and network infrastructure fail, and it carries local content that is never written down; it is also the dominant mass medium in many low-connectivity regions. Its main

failure mode is ephemerality, since unrecorded broadcast is unrecoverable.

17. Podcasts contain unusually candid and specific voluntary disclosure by insiders in a format with no editorial process. A statement in a podcast should be rated materially lower than the same statement in a filing, because a filing is compelled, formatted, and penalized, whereas conversational speech is unvetted, may be rounded or misremembered, and lacks review.

18. Fix the location by coordinates; establish the capture date of every image; compare the same frame across dates; describe changes in observable terms and separate the interpretation into its own sentence with a confidence level; corroborate the interpretation from another category.

19. Metadata is data a file or system records about content, generated automatically. It can establish creation and modification times, capture location, device, authoring software, authorship, editing duration, and relationships between files. It is corroborating rather than conclusive because fields can be edited, platforms strip them, ordinary handling such as re-saving or converting rewrites them, and device clocks are frequently wrong; absence is ambiguous.

20. Because items within one category share a production process and its failure modes, and may share an origin. Three news articles agreeing may be one wire story; a permit, a satellite image, and a local news report agreeing cannot be cheaply coordinated and fail in unrelated ways.

10.2 Multiple Choice Key

#	Answer	Note
1	b	A delivery channel, not a category
2	c	Production process
3	b	Secondary: it describes a primary document
4	c	A recorded deed is an original official record
5	c	Unindexed ordinary internet content
6	b	Allegations only
7	b	Recorder's deed records
8	b	One source
9	b	Misreading a field definition
10	b	Footprints, materials, and named occupancy at intervals
11	b	A distinct category producing a dated sequence
12	c	Corroborating, not conclusive
13	b	Monitoring services and liaison;

#	Answer	Note
		direct collection only under authorization
14	b	Three different categories with unrelated failure modes
15	c	A statement about the record system, not the world

10.3 True or False Key

#	Answer	If false, why
1	False	The deep web is unindexed ordinary internet content such as databases and case systems; the dark web requires overlay network software
2	False	A complaint contains allegations only; findings come from orders and opinions
3	True	
4	False	Absence reflects the system's jurisdiction, scope, digitization, and naming conventions, not the absence of the event
5	False	Most litigation occurs in state trial courts, and state trial court access is the weakest part of the system
6	False	Consumer platform base imagery may be years old, and its capture date is not always displayed
7	True	
8	True	
9	False	Within-category items share production process and failure modes and may share an origin; cross-category corroboration is substantially stronger
10	False	A settlement ends a dispute by agreement and establishes no facts or findings

10.4 Critical Thinking Guidance

7.1 Categories: corporate filings (registries in each claimed state, plus entity expansion), public databases (federal or state facility and licensing databases showing operating status), government records (permits and inspection records at each site), satellite imagery (physical confirmation and change detection at each site). "At least five years" of operation is verifiable through registration, permit, and imagery history. The claim open sources cannot fully verify is "similar," because comparability is a technical judgment about scale, technology, and configuration that would require engineering review of documents not necessarily public.

7.2 Categories: nonprofit information returns (the single highest-value document, because it is a compelled tax filing disclosing revenue, compensation, largest vendors, grants, and related entities under penalty), government records (grant and contract awards, licensing), court records, archived websites (prior program-scale claims), and newspapers. The information return is highest value because process: compelled, penalized, public, and structured.

7.3 Reasonable plan: maps (jurisdiction), government records (occupancy permit, business license, code enforcement, fire inspection history), public databases (safety and environmental inspections at the address), satellite imagery (recent and prior, for vehicles, materials, activity), and internet plus archived websites (current and prior claims, job postings for the location). "Unresolved" is correct if physical indicators are ambiguous, permits are silent or stale, and no inspection or hiring signal exists within the relevant period; students should state that explicitly rather than forcing a determination.

7.4 Answerable through: fire insurance maps, city directories, historical aerial photography, newspaper archives, and county permit and assessment records of the period. Not answerable through modern environmental or business databases, because those systems began decades later; their scope postdates the question, so a modern-database answer measures the reporting regime rather than 1955.

7.5 Audited filing: reliability A to B (compelled, audited, penalized), credibility 1 to 2. Blog post citing unnamed internal figures: reliability E to F, credibility 4 to 6. Product sentence: "The company's audited annual filing reports revenue of X (A1). A widely shared industry blog post asserts a different figure attributed to unnamed internal sources (F4); we assess the filing figure as the reliable value and note the discrepancy without adopting the blog's figure."

7.6 The outlet may be highly reliable (A or B) on the basis of its correction practice, editorial process, and track record, while the specific claim is low credibility (3 or 4) because it is unsourced, unverifiable by the reader, and unreproducible. The two are separate judgments, and the correct handling is to report the claim with its attribution and rating rather than either adopting or dismissing it.

7.7 Checks: whether it is peer reviewed (it is not), the venue and whether a peer-reviewed version has since appeared, sample size and methodology, whether independent replication exists, whether any expert commentary has critiqued it, funding and conflicts, and retraction status. Wording: "A preprint, not peer reviewed, reports X. The finding has not been independently replicated and should be treated as preliminary."

7.8 Partly reliable. The production process includes some obligations (publication requirements, potential audit, possible penalties for false official statements) but the agency has an interest in the conclusion, which is exactly the condition that lowers credibility for self-evaluative claims. The corroborating source is an independent audit or inspector general report, and secondarily the underlying program data if published.

7.9 No earlier appearance in reverse search supports the possibility that the content is new but does not establish it, because search coverage is incomplete and reverse search fails on cropped, mirrored, or re-encoded content. Absent metadata supports nothing at all, because platforms strip metadata routinely. Identical wording from four recent accounts supports coordinated amplification of the claim, and says nothing about whether the underlying video is authentic. The correct conclusion is that authenticity is unresolved and the amplification pattern is a separate finding.

7.10 Archived websites, compared systematically across captures for the product line's presence in navigation, product pages, and linked documents. Dating precision is limited to the interval between the last capture showing it and the first capture not showing it, expressed as a bracketed window. Corroboration from other categories (trademark filings, a discontinued-product notice, trade press, job postings) may narrow it.

7.11 A biased primary source is reliable evidence of what was published, in what terms, by whom, on that date, and therefore of the contemporaneous public framing of the dispute. It is not reliable evidence of the facts of the dispute. The professional handling is to cite it for the former and to seek other categories (court records, government labor filings, company records) for the latter.

7.12 One side's support collapses to a single origin, so its evidentiary weight is that of one report regardless of how many articles carry it. The other side rests on two independent government datasets, which is genuine cross-source support. The failure mode at risk is circular reporting mistaken for corroboration, and secondarily availability bias if the analyst counts articles rather than origins.

7.13 Three corroborating sources: a building permit for the structure (government records), a procurement or contract award referencing the construction (financial records or public databases), and a job posting or press mention referencing the new capacity (internet or social media). The permit would most raise confidence, because it is a compelled record naming scope, valuation, and inspection status, and its production process is unrelated to imagery interpretation.

7.14 Four sources across categories: corporate registry filings listing the individual as an officer or manager (corporate filings); property records showing the individual or a related entity as owner of the operating address (government records); a court filing naming both the individual and the entity as parties (court records); and a public statement, filing signature, or professional biography connecting them (internet, podcasts, filings). A sufficient second identifier is a shared registration number, a matching address on both records, a signature on a filing, a case in which both appear as parties, or an officer identification number where the registry issues one. Until one exists, the correct language is "a party of the same name."

7.15 Attempt corroboration by: testing internal details of the document against records (does the entity name, date, address, or figure match a filing or permit); checking whether the document's format and

template match other documents known to come from that organization; searching archives for any prior public appearance of the document; and checking whether the metadata is internally consistent and consistent with the claimed authoring environment. If all corroboration fails, conclude that the document's claimed provenance is unsupported and make no judgment about its authenticity. Wording: "An uncorroborated document of unestablished provenance asserts X. We have been unable to authenticate it, its metadata is inconsistent with the claimed origin, and we make no assessment of its accuracy."

7.16 The permit exists because the county requires one before construction may lawfully proceed, it is filed on a defined form describing scope and valuation, an inspector verifies the work at defined stages, and it becomes part of a public record maintained by a body with no stake in the project. The press release is written by a party whose purpose is to present the project favorably, is not filed under any obligation, has no defined content, and is reviewed by nobody outside the company. The difference is the process each document passes through, not the honesty of anyone involved.

7.17 The podcast statement is useful because it is specific, voluntary, and often more detailed than official disclosure, but it should be rated lower because it is unvetted conversational speech, may be rounded or misremembered, and carries no penalty for inaccuracy. It becomes the more valuable of the two when the filing does not contain the figure at all, when it postdates the most recent filing, or when it reveals a forward-looking dependency (such as a financing structure or a pending transaction) that periodic disclosure would not yet capture.

7.18 The production-process framework applies to community-contributed maps the same as anything else: contributions are logged with editor identity and timestamp, changes are reviewable and revertible, and heavily used areas receive substantial peer correction, which is a real if informal process. They are appropriate for feature detail in well-mapped areas, for rapid disaster mapping where no alternative exists, and as a corroborating source. They are inappropriate as the authoritative record of a boundary, an ownership interest, or any legally operative fact, where the definitive record is a survey, a legal description, or an official cadastral record. Check the edit history of the specific feature you rely on.

Appendix A: Reusable Templates

A.1 Source Selection Worksheet

QUESTION: _____
CONSUMER: _____ DEADLINE: _____
WHAT KIND OF THING IS THE ANSWER? (legal / physical / financial /
statement / historical / general-rate / technical / prior-claim /
provenance) _____
CATEGORY 1: _____ SYSTEM: _____
WHY THIS CATEGORY: _____
WOULD SETTLE: _____
CATEGORY 2: _____ SYSTEM: _____
WHY THIS CATEGORY: _____
WOULD SETTLE: _____

CATEGORY TO AVOID AND WHY: _____
CANNOT BE ANSWERED FROM OPEN SOURCES: _____

A.2 Collection Log

#	Claim	Categ ory	Sourc e / syste m	URL or citatio n	Acces s date/ti me (TZ)	Archi ved	Indepe ndent?	Rel.	Cred.	Verifi cation perfor med	Cover age bound ary	Analy st
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A.3 Cross-Category Corroboration Matrix

Claim	Supporting items (by category)	Contradictin g items (by category)	Distinct categories supporting	Collapses on inspection?	Determinatio n	Confidence + reason
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A.4 Negative Finding Template

Searched: [system name], [access method]
Query: [exact terms, including all name variants tried]
Date/time of search: [with time zone]
Result: no records returned
Coverage boundary: this system covers [jurisdiction], [date range],
[entity types]. It does not cover [what it excludes].
Therefore this finding does NOT establish: [the broader claim]
Next step to close the gap: [system or category]

A.5 Six-Property Source Assessment

Property	Assessment for this source
Content	
Production process	
Access model	
Coverage	
Latency	
Failure mode	

A.6 Evidentiary Characterization Reference (court records)

Document	Correct characterization	Never write
Complaint	"The complaint alleges..."	"Records show the company did..."

Document	Correct characterization	Never write
Answer	"The defendant denies..."	"The company admits..."
Motion	"In a filing, the party argues..."	"It was established that..."
Exhibit	"An exhibit filed in the case is a [document] that states..."	"Internal documents prove..."
Deposition	"In sworn testimony, [role] stated..."	"It was admitted that..."
Order	"The court found..." / "The court denied..."	(this is the one place "found" is correct)
Denial of summary judgment	"The court found material facts in dispute"	"The court ruled against the company"
Settlement / stipulated dismissal	"The case was resolved by settlement; terms were not filed. No findings were entered."	"The company settled, indicating liability"

Appendix B: Ethics and Self-Assessment Checklist

Confirm all twenty before submitting any product in this chapter.

Authorization and access

1. Every source was lawfully accessible to any member of the public.
2. I did not misrepresent my identity anywhere.
3. I did not defeat any access control, paywall, or credential.
4. I did not accept material from anyone unauthorized to share it.
5. I did not access the dark web.
6. I complied with terms of service where they applied.

Individuals 7. No private individual is a subject of my collection. 8. Individuals appear only in a professional or official capacity relevant to a stated requirement. 9. No minors appear anywhere. 10. No family members or associates of any subject were researched. 11. I created no aggregate profile of any person. 12. My ethics annex names every individual encountered and how each was handled.

Evidentiary discipline 13. Every fact carries its category, source, access time, and ratings. 14. Every negative finding carries its coverage boundary. 15. Allegations, exhibits, testimony, orders, and settlements are each characterized correctly. 16. No name match is asserted as an identity match without a second identifier. 17. Every date I assert is supported; bracketed changes are stated as windows.

Analytic discipline 18. My first sentence contains a judgment and a confidence level. 19. Facts are

visibly separated from assessments, and each significant finding states how many distinct categories support it. 20. I applied the same standard to claims that favor and disfavor every party.

Appendix C: Where to Go Next

Skills to build after this chapter, in rough priority order:

1. **Your local records landscape.** Spend an afternoon identifying, for your own county and state, the specific portals for assessor, recorder, permits, code enforcement, courts, corporate registry, UCC, professional licensing, and procurement. Bookmark them. This single exercise will make you more effective than any tool.
2. **Court records fluency.** Learn to read a docket, retrieve an exhibit, and characterize each document type correctly. This is the highest-leverage records skill in the discipline.
3. **Corporate structure research across jurisdictions,** including former names, trade names, and beneficial ownership registries where they exist.
4. **Geolocation and chronolocation** as practiced skills, using historical imagery for drills.
5. **Historical research methods:** fire insurance maps, city directories, historical aerials, and newspaper archive search strategies.
6. **Metadata handling,** including extraction, interpretation, and the defensive discipline of stripping it before publication.
7. **Records request practice.** Write and file one real public records request for something you are curious about. The experience of drafting a specific request and reading a response with exemptions applied teaches more than any description of the process.
8. **Data protection and computer misuse law in your jurisdiction.** The least glamorous and most consequential study available in this field.

Habits to start now:

- Keep a source log for everything, including casual research. The reflex is the skill.
 - For every negative result, write the coverage boundary. Every time, until it is automatic.
 - Archive at collection.
 - Once a week, take one claim you encounter in ordinary life and walk it back to its primary source. Most cannot be walked back, and noticing that is the point.
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End of Chapter 3